



Working among us

A Framework For A Healthy Environment

"The work that NEA does is important. It affects our lives on a daily basis. NEA as an agency is fantastic. They have people who know how to clean up, they know how to make sure that you can control dengue, how to manage hawker centres, they are very good. These are the silent workers."

— Dr Yaacob Ibrahim, Minister for the Environment and Water Resources (2004 - 2011)



NEA's work touches us in almost every aspect – the clean streets we walk on, the hygienic food establishments in which we dine and the healthy environment we live in.

The agency's officers walk and work among us, covering the ground to inspect, survey, enforce and even educate. Its team of researchers, analysts and officers from various fields support, improve and strategise NEA's operations, for the cause of better environmental and public health.

A sound legislative framework provides the foundation for the agency's operations. A number of relevant laws are administered by NEA, such as the Control of Vectors and Pesticides Act, Sale of Food Act, Infectious Diseases Act and Smoking (Prohibition in Certain Places) Act.

The backbone of NEA's environmental health work is the Environmental Public Health Act. Drafted in 1968 and implemented in 1969, the Environmental Public Health Act was a milestone in the history of Singapore's environmental health. Based on similar health acts in other countries such as Britain, the act encompassed comprehensive regulations on issues ranging from the management of food establishments and public cleaning services to vector control and hygiene in funeral parlours. The act brought Singapore up-to-date in addressing public health issues caused by rapid urbanisation and a burgeoning population.

Over time, various acts have been updated to ensure that they stay relevant and adequate to ensure a liveable, healthy environment. For example, regulations pertaining to food hygiene have been updated as recently as 2011 to incorporate higher standards for catered food.

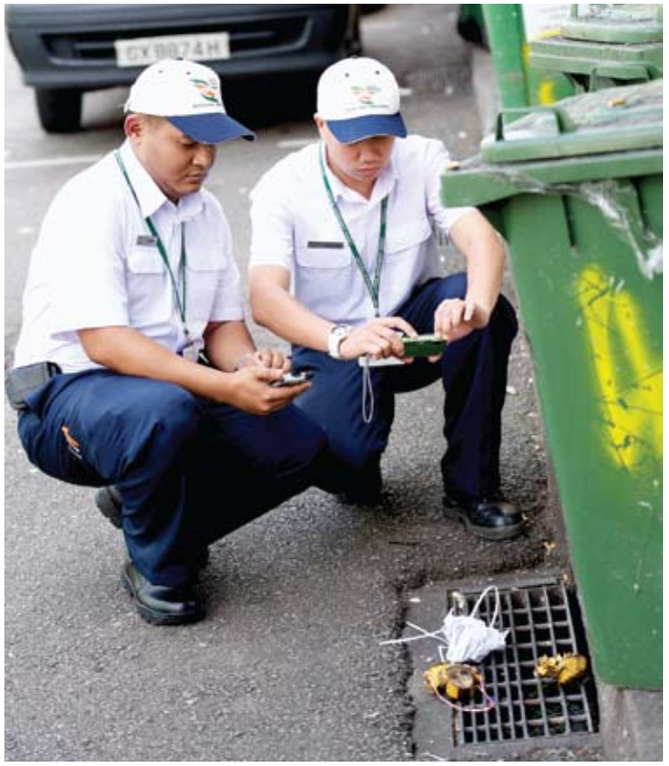
Accompanying these laws are Codes of Practice on various aspects of environmental health. Compiled for industry stakeholders, these codes comprise requirements, guidelines and best practices. For example, the Code of Practice on Environmental Health spells out legal requirements to address environmental health concerns in the design of buildings. Before the premises are allowed to operate, NEA will ensure that guidelines are complied with.

These codes of practice, together with the environmental laws, form the bedrock for NEA's work in regulating environmental public health and hygiene. They give the agency the authority to execute its work, and the mandate and mission to keep us in the pink of health.

Opposite: Mosquitoes can breed in stagnant water trapped in a space as small as this keyhole. Here, NEA officer extracts a sample using a pipette to analyse if breeding is taking place.



A Clean Act



Despite Singapore's clean image, litter still afflicts our environment. At its best, litter is unsightly. At its worst, litter can provide hospitable habitats for the breeding of mosquitoes and food waste for vermin, as well as pollute and choke our waterways. NEA manages Singapore's public cleaning regime, educates the public on anti-littering and enforces against litterbugs to ensure cleanliness in our environment.

A THOROUGH REGIME

Singapore's reputation as a clean city is in part a result of its thorough public cleaning regime, which is undertaken by private companies and NEA's daily-rated workers.

In the past, public cleaning duties were divided among various government agencies. For instance, public parks were taken care of by the National Parks Board (NParks) while vacant state lands were managed by the Singapore Land Authority (SLA). In April 2012, NEA began to progressively take over all the duties for public cleaning except for housing estates, which would continue to be managed by town councils. The move to integrate public cleaning services under NEA brought greater operational efficiency while providing a single feedback channel for the public on cleanliness-related issues.

GETTING TO THE ROOT

A reliable cleaning service alone is insufficient to keep our public areas clean. To get to the root of the problem, NEA uses public outreach campaigns to change mindsets and enforcement to deter and punish offenders.

The first national programme against littering was the month-long "Keep Singapore Clean" campaign launched in 1968. Annual campaigns to drive home the message became a regular feature from then on. After NEA's formation, it increased efforts to battle littering. In 2002, the "Singapore, Litter-free" campaign was launched, with an emphasis on ownership by the community to keep the country clean. The campaign went high-profile when mass events including National Day were targeted to be litter-free. In 2002, schools joined the campaign and in 2006, the campaign was extended to hawker centres. By 2008, coffee shops, bus interchanges and reservoir parks were roped in to extend the message's reach.

To deter would-be litterbugs, uniformed NEA officers patrol public areas regularly. The officers, together with auxiliary police, stake out littering hotspots to catch offenders. When a litterbug is caught, a ticket will be issued on the spot, requiring the offender to either pay a fine or attend court. To reform and deter littering behaviour, penalties such as fines and Corrective Work Orders are meted out depending on the severity of the act. These penalties apply to foreigners too.

"Environmental health is a highly challenging, dynamic field, particularly in a high-density compact city-state such as Singapore. In every aspect of our work, from ensuring high standards of food hygiene and vector control to keeping our public areas clean, from managing hawker centres and markets to running the country's only active cemetery, we seek to strike a balance between individual concerns and the greater welfare of the community, whilst keeping an eye on delivering services in the most cost-effective manner so as to meet public expectations without compromising on operational needs. Ultimately, we need the active co-ownership and participation of every citizen to help make Singapore a clean, healthy and liveable home for all of us."
— Derek Ho, Director-General, Environmental Public Health Division, NEA



REFRESHING THE ANTI-LITTERING CAMPAIGNS

Bins placed in more accessible and strategic locations, mothers playing a key role in spreading the anti-littering message and litter-free ambassadors shaping a new social norm of denouncing littering behaviour. These measures are part of NEA's renewed efforts to stamp out littering following two key studies on the issue.

Despite years of campaigns and strict enforcement, the problem of littering still persists. To understand what motivates such behaviour and how it can be addressed more effectively, NEA commissioned two important studies in 2006 and 2009 respectively.

The 2006 littering behaviour study saw researchers stationed in 203 public locations, from beaches, bus terminals and void decks to foreign worker congregation sites. They observed litterbugs and people who binned their rubbish, and conducted 3,018 face-to-face interviews with both groups of people. The key findings? Two out of 10 Singaporeans responded that they would litter and 83% of respondents felt that imposing fines would deter littering. The study led NEA to place more emphasis on engaging the community to change their mindset on littering.

In 2009, three sociologists from the National University of Singapore and a World Health Organization (WHO) consultant were brought in to conduct a more in-depth study of this issue. The panel researched and reviewed different aspects of littering, such as the profile of typical offenders, public perception of littering, communication strategies, enforcement procedures and the various social, cultural and environmental factors linked to the act. Focus group discussions and detailed interviews were held, and researchers surveyed 1,500 secondary school students and 3,000 households.

The findings were highly insightful. For example, major factors found to motivate littering behaviour include the lack of visible enforcement and bins nearby. Older people are more apologetic for littering than youths, women are more concerned with being caught littering than men and mothers feel that they should be positive role models for their children by not littering. NEA's anti-littering campaign was reviewed and found to be lacking specific messages to different target groups such as youths and smokers. The punitive measure against littering was also under-emphasised.

Armed with these insights, NEA refreshed its campaign again in June 2010. It improved the litter bin infrastructure, increased the frequency and visibility of uniformed enforcement and came up with more targeted public outreach and education efforts. More focus was placed on reaching out to mothers and youths as well.

Four months after the new measures were rolled out, NEA reviewed the effectiveness of the campaign. Visual checks on 20 littering hotspots revealed that the amount of litter had decreased. The number of litterbugs caught dropped from 41,392 in 2009 to 23,898 in 2010. By 2011, it had dipped further to 11,131. Most likely, this is due to the new anti-littering campaign in June 2010 and the deterrent effect of uniformed enforcement, which was carried out daily in 2011.



1967 - more than 1,000 people took up brooms and went on a "Sweep-Clean" campaign in Jalan Besar initiated by the Ministry of Health. *(Source: Ministry of Information, Communications and the Arts, Courtesy of National Archives of Singapore)*

SWEEPING THROUGH TIME

Alleys cluttered with discarded objects, sidewalks littered with piles of sweepings and drains clogged with garbage. Such was the normal state of affairs in Singapore even in the 1960s. Cleanliness in the city was a tall order when public cleaning operations were encumbered by the shortage of manpower and resources. Good social habits were generally lacking and many premises did not even have a dustbin. Dumping the sweepings from one's home or shop onto the streets was quite a standard practice then.

Shortly after Singapore became independent in 1965, the new government began to tackle the problem of public cleanliness. Harsher penalties for littering were imposed, dustbins were provided to households without one, public campaigns were launched and street patrols were organised to catch those who dumped rubbish illegally. In 1968, a programme was put in place for streets to be swept and refuse to be removed every day. The work was carried out manually by daily-rated employees. However, manpower shortage being an issue, the government introduced mechanical sweepers in 1972 for more efficiency. By the 1980s, much headway had been achieved as streets became cleaner and the resettlement of the majority of Singaporeans into Housing and Development Board (HDB) flats made domestic waste collection simpler.

To make the business of keeping the streets clear of rubbish more efficient, the government began outsourcing public cleaning services. By 2004, street cleaning services for the western, eastern and central parts of Singapore had been outsourced. Today, NEA carries out daily audits of the cleaning companies' services to ensure their reliability.

REFORM IN ACTION

You may have seen these people in bright, luminous green vests bearing the words "Corrective Work Order" sweeping away. Corrective Work Order (CWO) was introduced in 1992 to make litterbugs participate in public cleaning for a period of three hours, in order that they would understand the impact of their acts and the difficulties faced by cleaners. The clearly marked vests also identify them as litterbugs to distinguish them from cleaners, so that the shame of being punished in such a public manner would serve as a deterrent to potential litterers.



NABBING HIGH-RISE LITTERBUGS

It seems almost unbelievable that there are people who would throw sanitary pads, food waste and even soiled tissue out of their flats in high-rise buildings, but it does happen. To tackle such behaviour, NEA distributes educational materials and displays banners and posters regularly in the public to highlight the dangers of high-rise littering and to encourage greater social responsibility. When these acts take place, enforcement action requires evidence or statements by witnesses. NEA first serves letters to residents in the suspected unit and those living in the same column of flats to advise against high-rise littering. Warning letters come next when such acts do not stop. And if the behaviour persists, the agency will proceed with stake-outs to gather evidence against the culprits, followed by prosecution. NEA will also be rolling out surveillance cameras for high-rise littering hotspots to aid in nabbing such offenders after a successful pilot in 2011.





"In any policy review, we need to first understand the needs and concerns of all affected parties and find a balance between meeting public expectations and ensuring that the policies and measures introduced are practical and implementable in addressing those needs. We will focus on areas where the public will be impacted by second-hand smoke in their daily lives and at community spaces where human traffic is high. We also take into consideration spaces used by vulnerable groups; hospital compounds, nursing homes, children's playground, etc. Our long-term aim is to allow smoking only at designated areas. Hence, transiting from a prohibitive list to a positive list which states places where smoking is allowed. Think about it, as a citizen, what would you expect? You would expect that the moment you step out of your home whether to work or play, the authorities would ensure that you do not have to pass through second-hand smoke if you do not wish to. When you go to bus stops or train stations or visit any food places, you wish you have a choice to say, 'I don't want to go through a smokescreen'. Together with the support from the public and stakeholders, we will continue to increase the protection against second-hand smoke by making more public places smoke-free. We hope that the community too plays its part, for example by politely advising smokers to refrain from smoking at prohibited places and encouraging them to avoid lighting up where second-hand smoke could impact non-smokers. The environment is for all to enjoy and each of us has an important role to play."

– *Chua Yew Peng, Director, Policy and Planning Division, NEA*

OUT WITH SMOKE!

Now you can enjoy an al-fresco dinner, groove the night away at one of Singapore's many nightlife spots, and work out at designated exercise areas, without having to suffer second-hand cigarette smoke. These are just some of the areas which became smoke-free after 2006 when the Smoking (Prohibition in Certain Places) Act was extended.

The government has been progressively increasing the number of smoke-free public places since 1970 so that there are more spaces where the public may use or enjoy without having to put up with the hazards of cigarette smoke. The prohibition kicked off with omnibuses, theatres and cinemas and was gradually extended over the years to virtually all indoor places and many outdoor community spaces.



The prohibition of smoking in public places is part of the government's National Tobacco Control Programme, led by the Health Promotion Board and supported by agencies including the Health Sciences Authority, Singapore Customs and NEA. The programme aims to reduce smoking prevalence in Singapore by adopting a holistic approach including public education, legislation, taxation and provision of smoking cessation services.

The smoking prohibition policy adopts an integrated approach comprising policy or legislation review, enforcement and public communications. Consultations with the public and stakeholders are carried out to gather public sentiments and address operational concerns. For example, for the extension of the ban to hawker centres and food establishments on 1 July 2006, and entertainment outlets on 1 July 2007, NEA had begun the public consultation process in 2005 through an e-poll. In addition, focus group discussions were held with smokers and non-smokers, and dialogue sessions were carried out with owners of food establishments and entertainment outlets. This enabled NEA to fine-tune the policy to meet public needs while balancing it with operational considerations of stakeholders, thereby ensuring a smooth roll-out of the extension.

The result? While hawker centres, food establishments and entertainment outlets were made smoke-free, provisions were given in the law for operators to set up designated smoking areas so as to cater to patrons who were smokers. The early announcements and engagement efforts gave the operators ample time to prepare for the ban.

In addition, public communications and education efforts played an important role in preparing the ground for the implementation of the policy. Media publicity efforts, guidebooks for premises operators, advertisements and an educational blitz were launched to ensure that the public and stakeholders were fully aware of the upcoming ban. Enforcement was also carried out to ensure compliance and deter smokers from flouting the laws.

The same approach was again adopted for the next smoking ban extension announced in March 2012. Over 8,000 responses were received through the e-poll survey conducted between November and December 2011, with 89% supporting the call for more smoke-free places. Feedback was also garnered on the places where the public would like to be smoke-free. Beyond the e-poll, NEA also collected feedback through social media platforms, discussion forums and feedback channels. Consultations with stakeholders including HDB, LTA, town councils and other premises owners and managers were also conducted. Such exercises allowed NEA to work out the details of implementing the ban's extension with the different stakeholders.



SLICK OPERATIONS

When it comes to cleaning the environment, it's not just about streets and pavements. On 25 May 2010, an oil tanker and a bulk carrier collided in the Singapore Straits just 13 km off Changi East. Some 2,500 tonnes of crude oil from the tanker spilled into the sea.

The Maritime and Port Authority of Singapore (MPA) formed the first line of defence. It moved in immediately to contain and treat the spill. NEA, on the other hand, began mobilising resources to clean up the shores should the oil patches spread. Its air and water quality monitoring teams were despatched on the very same day to assess the impact of the spill on the environment. The teams found the levels of toxic substances in the air and water to be within safe limits.

The next morning, oil patches were detected off Changi Naval Base and a nearby rock bund. By the afternoon, the patches had spread to East Coast Park beach. Within the next few days, Tanah Merah and Chek Jawa at Pulau Ubin were hit too.

NEA straightaway closed East Coast Park and Changi beaches to the public and launched an intensive clean-up operation. Some 30 NEA officers were deployed to conduct surveillance at different stretches along the eastern coast. About 500 personnel from NEA, a cleaning company and employees of the tanker owner took part in the operations.

From 7 am to 7 pm every day, these personnel worked to clean up the patches. Dispersants were used to treat the oil, and contaminated sand was removed and disposed of. Operations sped up after NEA successfully tested a more effective biodegradable agent comprising nanoparticles to clean thick patches of oil at East Coast and Pulau Ubin.

At the same time, NEA led other government agencies and volunteers to play different roles in the operation. Daily meetings were held to coordinate ground operations and bring everyone involved up to speed. At Chek Jawa, oil absorbent booms and padding were put up to keep the oil out of the area. Patches that seeped through were cleaned by NEA, NParks and volunteers. Thanks to the swift response, no major damage was done to the eco-system.

PUB, the national water agency, installed oil absorbent float booms to prevent oil patches from entering drains and canals along East Coast Park beach. It also brought in a vacuum tanker to suck up oil scum from the water surface.

The rapid response and collaborative efforts facilitated a faster, smoother clean-up act. Ten days after the operation took off, the beaches were declared open to the public on 4 June 2010.



"Even as Singapore develops and progresses, and our population increases, we have to make sure that our environment stays clean, liveable and sustainable. It's not enough to just keep it at status quo, though that in itself will be a challenge since the stress placed on the environment will grow as the country develops further. We have to find ways to make our environment even better, cleaner and healthier. The responsibility for this falls upon MEWR, and its statutory board, NEA. Which is why over the years, NEA has consistently introduced measures and initiatives to improve our surroundings. The extension of the smoking ban is but one example. For many members of the public, the environment has become more pleasant as a result. Another important initiative is the consolidation of public cleaning services under NEA for more effective and efficient operations. As we move forward, NEA's work in maintaining food hygiene standards, vector control and research and public outreach will become all the more challenging and critical."

— Grace Fu, Senior Minister of State, Ministry of Information, Communications and the Arts, Ministry of the Environment and Water Resources

A Safe And Clean Food Paradise

NEA relies on stringent systems of licensing, enforcement and ground inspections to perform its duty. However, given the number of food operators and handlers in Singapore, continuous monitoring is impractical. At the end of the day, those in the food and beverage industry need to do their part too in taking care of hygiene and cleanliness.

Talk about Singapore being a food paradise. In 2011, there were a total of 5,855 cooked food stalls in government hawker centres, 13,829 food stalls in private eating establishments, 14,363 private food shops (including restaurants, fast food outlets, eating houses, food courts and caterers), and 367 supermarkets. Over 100,000 food handlers work in the industry, providing sustenance for Singaporeans and tourists alike.

These were also the numbers of food establishments NEA had to inspect and regulate in 2011 alone to ensure a high level of hygiene and cleanliness.

Food safety in Singapore comes under two agencies – NEA and the Agri-Food and Veterinary Authority (AVA). AVA regulates food safety at the stages of import, manufacturing, distribution and wholesale while NEA regulates retail food service establishments to ensure that food is prepared in a hygienic manner.

Maintaining high standards of food hygiene in Singapore is a task of herculean proportions. Every year, more and more food and beverage outlets open for business. With more locals dining out and increased tourist arrivals, the number of patrons at food establishments grows by the day as well. In 2004, a survey showed that 48.4% of Singaporeans ate out, a figure which rose to 60.1% in 2010. On top of that, there is also a great disparity in the standard of hygiene and food safety procedures across the food and beverage industry.

To make sure that this veritable food paradise offers safe and clean food at all times, NEA relies on stringent systems of licensing, enforcement and ground inspections. However, with the number of food operators and handlers in Singapore, continuous monitoring is impractical given our finite resources. At the end of the day, those in the food and beverage industry need to assume the primary responsibility in ensuring that food for sale is prepared under hygienic conditions. Consumers can also play their part by not patronising food outlets with poor hygiene practices.

A SYSTEM FOR TRACKING

It is just a piece of paper, but this licence makes all the difference in many ways. It allows food operators to operate legally and NEA to track their performance and records more efficiently. The system of licensing also aids NEA in working out inspection schedules and taking swift action in the case of any non-compliance.

Every retail food establishment, be it a stall in a hawker centre, a restaurant, or a fast food outlet, is licensed by NEA, subject to compliance with licensing requirements and the Code of Practice on Environmental Health. In the case of hawker centres and markets, NEA also manages the tenancies.

WATCH FOR THE DECAL

Grading is not just for students. At every hawker stall and food shop, you can clearly see a decal displaying the grade of the premises. These grades are given every year by NEA, whose officers assess the cleanliness and hygiene of the premises before giving the food establishments a grade.



"We take cleanliness seriously not just for the 'A' grade, but because people are eating our food, so we must take care of hygiene."

— Kwan Wai Kern, proprietor of Sin Tong Nam, a bakery with an 'A' grading at Tiong Bahru Market

Implemented in 1997, the system's highest grade is "A", when the premises score 85% or more on NEA's checklist. "D" is the lowest grade, when the score is between 40% - 49%. These grades are given based on criteria such as cleanliness of food preparation areas, implements and equipment and the hygiene practices of food handlers.

The grade must be displayed prominently so that the public can gauge the hygiene standards of the outlet. With pressure from both the authorities and rising public expectations, food hygiene standards have gradually improved over time. Back in 1997, less than 50% of food establishments scored "A" or "B", compared to more than 90% in 2011.

Moving forward, NEA will adopt a risk-based assessment framework in its food hygiene surveillance. Key factors such as the types of food, operating hours, clients and past records of operators would be taken into consideration in risk-profiling of food premises for inspections. For example, a restaurant selling raw food such as sushi runs a higher risk of food contamination while caterers serving vulnerable groups including the young, sick or elderly have to be monitored more closely. This system, to be used internally, will allow NEA to deploy its resources more efficiently.

INSPECTIONS, UNANNOUNCED

It's done on a regular basis, it's unannounced, and sometimes, officers in plain clothes make an appearance.

Inspections are part of NEA's efforts to ensure food hygiene. These surprise spot checks allow NEA to know what is happening on the ground and take action where necessary. On average, more than 12,000 inspections were conducted every month in 2011.

Should officers find any flouting of regulations, offenders will be penalised. The Points Demerit System, introduced in 1997, is used to assess the seriousness of the offences. Demerit points are given for lapses in hygiene, such as failing to keep the establishment free of vermin or insects, sweeping food scraps onto the ground and preparing food on the floor. Depending on the number of demerit points accumulated, the business may face suspension or may even have its licence revoked.

NEA officers also go around the island in plain clothes to purchase food from establishments randomly. After the purchase, the officers will reveal their identity to the sellers. The food is then tested for pathogens in accredited laboratories. Besides facilitating the collection of data on food contamination, these spot checks also help motivate sellers to uphold better practices in food hygiene and safety.

PUSHING FOR HIGHER STANDARDS

To push for higher standards of food hygiene in the industry, NEA requires all food handlers and operators to attend and pass relevant courses developed jointly by NEA and the Workforce Development Agency (WDA). Topics covered range from micro-organisms and food contamination to hygienic ways of preparing, handling and storing food. Every three years, food handlers must undergo a refresher course.

NEA also organises regular briefings and dialogue sessions to inform and educate stakeholders in the industry. At the end of the day, food establishments depend on the public for business, which means that public pressure is more likely to be effective. Therefore, NEA holds regular public health education workshops and talks to create more awareness among consumers on hygiene issues. It also briefs grassroots organisations and government agencies on safety guidelines on catering when festive seasons come round. Track records of licensed food caterers are published on NEA's website so that the public can avoid using unlicensed or unreliable caterers.

Opposite: Measuring temperature of cooked food to ascertain that the food has not been left standing for long. NEA officers also take the opportunity during inspections to educate operators on control of food temperature and serving time.


LAU WEI HAR
TAN AIK KEAT

"NEA has made critical contributions to Singapore's liveability over the years through its relentless efforts to uphold high standards of public health and hygiene. This is despite more economic activities and a larger population. It is well-positioned not only to continue in this excellent work, but also to lead in new areas such as climate science, environmental research, and industry development. Much of NEA's success can be attributed to its dedicated and professional staff who continue to seek new solutions to emerging challenges, and to respond swiftly and effectively when faced with new demands. Over the last 10 years, NEA has successfully achieved the operational effectiveness and service responsiveness which were the objectives of its establishment as a statutory board in 2002. It is a cohesive and resourceful organisation that works closely with MEWR and many other stakeholders in the public sector, industry and community."

— Rosa Daniel, Deputy Secretary, Policy, MEWR



THE PROBLEM OF FOOD CONTAMINATION

Have you ever left food in your home only to find it going bad or mould growing on it some time later? Singapore is warm and humid – ideal conditions for bacteria and mould to multiply and spread quickly as compared to countries with temperate or drier climates. These bacteria and mould can produce toxins that are harmful to health.

This is only one of the many challenges of preventing food poisoning in Singapore. Contamination can be caused by many different factors – such as poor personal hygiene, dirty premises, food left standing for too long, unclean cooking implements, contaminated ingredients, undercooked food or cross-contamination with other sources.

When food poisoning incidents do happen, the investigation falls primarily upon three agencies – AVA, the Ministry of Health (MOH) and NEA. AVA is involved if the food source is suspected to be contaminated, such as the food manufacturer or industrial kitchen. MOH investigates which bacteria or toxins are responsible. NEA checks the retail outlet to verify what has gone wrong.

Such investigations are highly challenging processes. As the food and beverage industry is fast-moving, inventory is usually exhausted by the end of the day, sometimes even before afflicted consumers report falling sick. Generally, food businesses also tidy up and clean their premises when the day is over, which makes it hard for investigators to find the source of contamination. For example, if a chopping board cross-contaminated with bacteria is responsible for a outbreak, it will usually be cleaned by the time food poisoning was reported. Over at MOH, ascertaining the cause of poisoning takes time as well. Patients generally consume a wide range of foods from different sources every day, and pinpointing the responsible source is a challenge if the outbreak is not massive. Investigation thus takes considerable time and it is often hard to identify the exact cause of poisoning in many cases.

Despite all of the government's efforts to regulate food safety and hygiene, food poisoning cannot be eradicated completely. Just one hygiene lapse, such as handling cooked food with unwashed bare hands, is enough to cause an outbreak. Ultimately, food handlers and operators are responsible for the safety and health of the food they sell.

"NEA's strength lies in its team of dedicated and professional staff. They are a cohesive team and are very knowledgeable and experienced in their work. It was indeed an honor and privilege for me to have worked with them. They are very committed to their jobs, the unsung heroes of our clean, green and healthy environment. They carry out their duties and responsibilities untiringly, without fear or favour, be it in enforcing against littering, enforcing the smoking ban, sweeping the streets, checking on hawkers, fighting the dengue scourge, operating the incineration plants efficiently, checking on construction noise pollution and so on. They work behind the scenes and are seldom in the limelight. When things are all right, their efforts go unnoticed and often unappreciated, but when things go wrong, they are cast into the limelight. It is important that NEA continues to attract and retain a fair share of the nation's talents and develop a core team of professional officers with the passion for the environment."

— Lee Yuen Hee, CEO, NEA (2005 - 2009)

STAMP THE TIME!

Before you head for the catered buffet line or dig into a packet of catered food, check the time-stamp of the food. If it has been more than four hours since the time-stamp, avoid the food for your health's sake.

Since 15 February 2012, NEA has required all food caterers to time-stamp all buffet and packed meals. A time-stamp sign must be displayed prominently on buffet tables set up by caterers. Catered packed food should also have labels indicating the time-stamp. The agency will take action against those who do not time-stamp their food, or who provide false or misleading times.

This is because when cooked food is stored between 5°C and 60°C, bacteria may multiply quickly to unsafe levels or produce toxins which can cause food poisoning. To prevent this from happening, cooked food should be eaten within four hours if it is left standing within this temperature range. However, consuming the item within four hours is not a guarantee that food poisoning will not occur. Undercooked food and contamination caused by unhygienic preparation, dirty food service equipment and poor handling may still cause food poisoning.

Catered food poses higher risks to consumers than dining in restaurants or hawker centres, because a large amount of food is prepared in advance and stored before it is transported to the venue where it may stand for a while before being consumed. Between 2009 and 2011, caterers accounted for 21 of 75 major food poisoning incidents and 35% of all persons affected in major food poisoning incidents, even though caterers comprise only 1% of the food establishments in Singapore. This led NEA to impose more stringent requirements on food caterers and conduct inspections more frequently. NEA also requires all food caterers to engage at least one certified Food Hygiene Officer to oversee and audit the hygiene standards and operations of the caterer.

The time-stamp requirement came about after consultation with the food catering industry. Educational materials on time-stamping are also made available to inform caterers of the requirement and rationale. Says James Wong, President of Association of Catering Professionals Singapore, "The new regulation can serve to protect both the industry and the consumers from food poisoning ... the association will continue to work with its members to improve on industry best practices to ensure safe food for all."



SAFE WATER TO DRINK

The quality of piped drinking water in Singapore is one of the safest in the world, and well within guidelines set by WHO. As an added measure to guarantee that our piped water is safe for drinking, NEA was appointed in 2007 as an independent auditor to regulate the quality of water supplied by PUB. Using guidelines developed by a technical committee comprising experts from fields such as toxicology, water treatment, medicine and microbiology, NEA conducts audits and visits the suppliers on-site to test the water quality for compliance with these guidelines.

WHEN HYGIENE IS UNDERMINED

4.30 *am*, 4 April 2009. MOH informed NEA of a food poisoning outbreak. 7.30 *am*, Kandang Kerbau Women's and Children's Hospital contacted NEA, reporting 57 cases of food poisoning cases. All the cases were linked to the Rojak Geylang Serai stall in the Geylang Serai temporary market. 8.00 *am*, NEA officers arrived at the suspected stall to conduct a joint investigation with MOH.

Samples and swabs were taken. The stall holder and four assistants were sent for screening for food-borne pathogens at Tan Tock Seng Hospital. Business operations of the stall were ordered to stop immediately. While NEA took charge of the scene, MOH focused on finding out the cause of the food poisoning. As an added precaution, NEA also conducted an inspection of all stalls at the temporary market.

It was the worst case of food poisoning in recent times, with two fatalities and 152 ill, 48 of whom were hospitalised. One victim suffered a miscarriage.

THE "RAT" HERRING

While investigations were ongoing, public speculation swirled. Just two days before the outbreak, a local paper had run a story on rodent infestation at Geylang Serai temporary market, accompanied by vivid images. Despite the lack of evidence, the media and online community leapt to the conclusion that rats were responsible for the outbreak. This was followed by fervent public debate on the government's role in curbing rodent infestation and maintaining hygiene standards.

THE CAUSE: *VIBRIO PARAHAEMOLYTICUS*

Over at MOH, laboratory investigations finally identified the bacterium behind the outbreak. It was *Vibrio parahaemolyticus*, a pathogen found in raw seafood and known to cause gastro-intestinal diseases. However, no one was able to ascertain how the bacteria got into the rojak. Before NEA and MOH arrived on the scene on 4 April, the food handlers had discarded all the remaining food and cleaned up their premises. Most likely, cross-contamination between raw seafood and the cooked food had taken place.

A REALLY UNHYGIENIC STALL

Although the authorities could not establish how the contamination had taken place, two things were clear – that the stall was the source of food poisoning, and that the food handlers practised poor hygiene. The chopping board used in the stall was found to be carrying faecal coliform, which comes from faeces, and *Staphylococcus aureus*, a bacterium is found on human skin. Swabs taken from various areas in the stall showed traces of *Salmonella*, a bacterium which causes food poisoning. The stall holder was subsequently fined and debarred from applying for a hawker or food shop licence and registering as a food handler. His assistants had to re-take and pass the Basic Food Hygiene Course again as a precaution.

ADDRESSING PUBLIC CONCERNS

Meanwhile, NEA had to address public speculation and debate on rats and hygiene. On 18 April 2009, an open letter from the CEO of NEA, supplemented by press interviews with Khoo Seow Poh, then Director-General of Public Health, was published in full in newspapers. In the letter, the CEO expressed his sympathy for the victims and provided assurance that NEA would continue to uphold high standards of food hygiene.

Even though rats did not cause the outbreak, the CEO acknowledged that NEA should have moved in earlier to help address the rodent infestation. During the crisis, NEA, together with the Geylang Serai Temporary Market Management Committee, had stepped in to cut down rodent population at the market.

A STRICTER HYGIENE REGIME

Despite the fact that the outbreak was caused by personal hygiene lapses, NEA decided to use this episode to further strengthen its regime on the control of food hygiene.

NEA stepped up its inspection of markets, food centres and food stalls, and increased fines for lapses. The mandatory Basic Food Hygiene Course for food handlers was revised to emphasise proper hygiene practices to prevent food contamination.

THE QUESTION OF TEMPORARY MARKETS

Though it was established later that the food poisoning outbreak was not caused by rodents, the question of cleanliness in temporary markets became a sticking point during the crisis.

When hawker centres are upgraded under NEA's Hawker Centres Upgrading Programme, temporary markets are built and maintained by the local Citizens' Consultative Committees, who usually form specific management committees to oversee the markets' operations.

After the Geylang Serai food poisoning incident, NEA now includes its officers on the management committees of temporary markets and works more closely with the committees to improve housekeeping practices. In addition, from 1 June 2009 onwards, all operators of temporary markets or food centres have to be licensed. They will be held liable for hygiene lapses so as to ensure accountability.

Off The Streets, Under New Roofs



A typical hawker stall in the 1970s. *(Source: The Straits Times © Singapore Press Holdings Limited. Reprinted with permission)*

Gone were the days when hawkers would file past your home, calling out their wares and ringing bells and such to attract customers on the streets. Instead of sitting out on rickety stools under the hot sun and eating your food off a plate of dubious cleanliness, you can now have your pick of meal in the comfort of well-ventilated, sheltered hawker centres.

Many of us remember the days of street hawking with nostalgia, often forgetting the poor hygiene that went along with it. Food waste was often dumped indiscriminately on the streets and in drains by hawkers and patrons, causing pollution and increasing the vector and vermin population. Hawkers would go around with two buckets of water to be used through the day for washing a small stack of plates for serving food. Such poor hygiene practices led to outbreaks of cholera, typhoid and dysentery. Street cleaning was made difficult as roads and drains were obstructed by hawkers and their props. In parts of the city where hawkers congregated, the areas would look haphazard as hawkers built makeshift structures to act as stalls. Those who sold goods other than cooked food spread their wares out to attract customers, congesting streets, alleys and even roads. It was hard to eradicate street hawking as many of the hawkers were hardship cases.

By the late 1960s, however, the rate of employment in Singapore grew and people became more affluent. Street hawking was no longer the main means of livelihood available. As such, the government began to control the upsurge of street hawking to eradicate the accompanying public health and environmental issues. An island-wide census of street hawkers was conducted between 1968 and 1969 and some 24,000 street hawkers were registered.

To solve the root of the problem, the government began building proper centres to house street hawkers. These centres had potable water supply, proper sanitation and sewage systems, refuse disposal facilities, electricity and ceiling fans installed for ventilation. HDB also started constructing hawker centres and markets as part of its facilities for residents in new public housing estates. By 1986, all street hawkers were finally resettled into the newly-built markets and hawker centres. It was a 15-year effort which cost $100 million.

The time and resources spent are well worth it. Today, hawkers are able to practise their livelihoods in well-equipped facilities. Public health problems arising from poor hygiene practices declined significantly with the resettlement of hawkers. Pollution of waterways and dirtying of streets due to street hawking is curbed. The roads are free from congestion by make-shift stalls and unregulated goods display. And Singaporeans get to have good, affordable, hygienic food in clean hawker centres, which also double as community spaces where friends and families go to socialise. Over time, hawker centres have evolved to become a well-loved national icon, and are even listed as one of the top places to see in the landmark book “1000 Places to See Before You Die”!

"We are exploring new not-for-profit management models, creative architecture and innovative designs that make the centres more pleasant without making it more expensive, which encourage social interaction for Singaporeans from all walks of life. It is formal recognition that hawker centres are an integral component of our social infrastructure."

— Dr Vivian Balakrishnan, Minister for the Environment and Water Resources



CATCHING UP WITH TIMES

If you could recall, just a decade ago, our hawker centres and markets were becoming run-down. Most of them were built 20 to 30 years ago, and the facilities could no longer keep up with changing needs and times. This becomes particularly noticeable as surrounding HDB blocks were gradually upgraded.

Regular maintenance of hawker centres and markets owned by MEWR is undertaken by NEA, while respective town councils oversee those owned by HDB. However, maintenance alone is not enough to ensure that these facilities remain serviceable.

Recognising the central place of hawker centres and markets in the lifestyle of Singaporeans, the government embarked on upgrading these facilities in 2001 with the launch of the Hawker Centres Upgrading Programme (HUP). Under the programme, $420 million was set aside to upgrade hawker centres over ten years. The first batch of centres to be upgraded was announced in July. The upgrading allowed hawkers to better compete with supermarkets and food courts and also increased the appeal of the centres.

The objective of the upgrading was to provide a more pleasant dining and marketing environment for stall holders and the public. Before every HUP project, a working committee would be formed, comprising different stakeholders such as grassroots members and hawker representatives, so that matters such as the upgrading schedule, layout of stalls and design of the centre could be discussed and decided upon. Feedback was also gathered from stall holders so that their concerns could be addressed.

The programme was a success. Dining and marketing in the newly-upgraded centres became more pleasant. Hygiene and cleanliness standards were easier to upkeep in the new buildings. Care was taken to preserve the charm and ambience of old hawker centres in the new structures while giving them a distinctive identity. Under the programme, a total of 107 hawker centres and markets would be upgraded by 2014.

As Singapore's population grows, it became clear that new hawker centres were needed in the future as more Singaporeans were eating out. In October 2011, Dr Vivian Balakrishnan announced plans to build new hawker centres, followed by the confirmation in 2012 that seven new centres would be constructed over the next five years.

The government set up a public consultation panel on the issue in December 2011 and invited members from social enterprises and different fields such as food and beverage, environmental sustainability and architecture to come up with ideas on how the new hawker centres should be. These moves ensured that Singaporeans have a stake and say in the making of our future hawker centres. More importantly, hawker centres would continue to be part of what makes Singapore unique.



THE SPECIAL SQUAD HITS

"There were a lot of complaints where people buy canned abalone at a special price. But when they went back, they open up, it's not what they get ... And then you have these people selling talcum powder ... adulterated the talcum powder with flour. They get this empty container from those people who collect them for sale. So they open up the back part of the container, it's easy to remove it. Take out the talcum powder, say about 80 percent, leave 20 percent on the top and then the rest you will (top) up with flour."

– Phillip Chew Peng Kia, who worked as a public health inspector between 1957 and 1974, on the goods sold by illegal hawkers (Source: Oral History Centre, National Archives of Singapore)

There is a long list of reasons why illegal hawking had to be eradicated. Street hawkers are sources of unwholesome food, fake and contraband goods, pollution and street obstruction. Singapore's efforts to end illegal hawking was carried out in fits and starts till 1973, when ENV began intensifying efforts to stem such activities. An operations room for the Hawkers Department was set up at Princess House. It served as a nerve centre to monitor via radio the movements of ENV's fleet of 21 radio-equipped vans, and to co-ordinate and direct the activities of field staff. Information, such as tip-offs on sites of illegal hawking, could be relayed quickly. Requests for back-ups could also be sent immediately.

Four inspectors from the Special Services Squad, accompanied by CISCO guards, were assigned to each van to patrol the streets. Each day, four or five teams would be sent on raids. During these raids, goods were seized, cooked food thrown away, and the offending lorries and cars impounded. Perishables and fresh produce were given to charities.

It was not an easy job. Most hawkers depended on hawking for their livelihoods and public sympathy was usually with them. There were cases where illegal hawkers turned aggressive and put up resistance during the raids. But by 1975, the situation came under control as over 18,000 street hawkers were relocated into new centres and markets.

Today, street hawkers are a rare sight in Singapore. Many of such hawkers today are part of syndicates and are involved in selling pirated, contraband or illegal goods. The Special Squad, as it is now known, continues under NEA with its mission to crack down on these activities.

A NEW LEASE OF LIFE

Remember the old, dank hawker centres and markets before HUP? Thanks to the programme, these old buildings were given a new lease of life.

Brighter surroundings, better ventilation, cleaner stalls and floors – these are the new features that stand out at first glance. But there's more. Depending on the state of condition the centres were in, different improvements were carried out. The standard list of upgrading features included re-tiling, replacement of tables and stools, re-wiring and replacement of exhaust systems, sewers and water pipes and improvements to toilets and bin centres.

If a hawker centre had an estimated lifespan of another 10 years after upgrading, it would then be reconfigured or rebuilt. This means bigger stalls, better facilities, fire safety features and even a new architectural design and layout. In some cases, hawker centres and markets located near each other were consolidated and rebuilt as a single larger centre, for example Commonwealth Crescent Market.

To facilitate smoother operations in the upgraded centres, central freezer and washing areas were built. Barrier-free access was incorporated to facilitate wheelchair-bound patrons.

As HUP was not run on a cost-recovery model, rental increments were moderate. Stall holders were also given several choices to decide if they want to continue with their livelihoods at the same centre or other non-upgraded centres, or to opt out with an ex-gratia payment package.

These cleaner, brighter, well-ventilated centres and markets were warmly received by the public. A survey commissioned by NEA and the Ministry of Community Development, Youth and Sports in 2005 showed that 92% of respondents felt that upgraded centres were cleaner and 68% said that they would visit the facilities more often.

What's more, hawker centres are also turning into vibrant spaces for the community as NEA works with grassroots organisations, residents and the hawkers to organise food fairs, live music and performances at these venues. So that the public can catch up on the latest buzz at these icons of Singapore, NEA has also launched a website, myhawkers.sg, in 2010 to serve as an information portal.

Newton food centre through time











Top left: Newton Food Centre in 1977 *(Source: The Straits Times © Singapore Press Holdings Limited. Reprinted with permission.)*
Middle, left: The food centre during the 1980s. *(Source: Ministry of Information, Communications and the Arts, Courtesy of National Archives of Singapore)*



A CULTURE OF SAFETY

Devin Otto Kimble, Managing Director of MENU Pte Ltd Food & Drinks Group, the company behind Brewerkz Restaurants and Microbreweries, set up shop in Singapore after stints in the food and beverage industry in Cambodia and Hong Kong. He shares with us his thoughts on food hygiene and NEA.

Of all places, why did you choose to set up business in Singapore?
Singapore is really at the crossroads of civilisations, you can breathe the air here, and the legal system is transparent. When we set up in Singapore in 1997, the main considerations were that the laws were very clear. We could find out what the regulations were for a food business easily.

As someone who is in the food and beverage industry, how would you characterise your relationship with NEA, who is a regulator?
We are all trying to get to the same place, to put safe food on the table. We get 1,000 or more customers a day, and if I start getting people sick, I can hurt a lot of people. So we are on board with NEA on this.

Sometimes, there are things we overlook, and we'd say, 'Argh, we should have done that.' Once, we wanted to move the location of our bar and the NEA officer rejected the plan. So we had to spend another $10,000 to get the architect to draw the plan again, and this time, making sure that sinks for hand-washing and such were included.

But it makes sense and it's incredibly important to have people who are looking after these issues. Because I'm an operator and I'm more concerned with having enough people to work on the floor and that they get their pay checks. So we can depend on NEA as the steward. Because if Singapore is known for making people sick, then people won't feel as comfortable here.

What about the implementation of designated smoking areas and the smoking ban? Did it affect your business in any way?
It didn't affect our business at all! I've always thought it was the right thing to do. I noticed when I first came down to Singapore from Hong Kong that I didn't have to send my jackets and things out for dry cleaning as much. Back in Hong Kong, the cigarette smoke in restaurants was awful so it was a joy to be able to work in a smoke-free environment.

Can you share with us some of the changes you feel NEA has made to this industry?
We used to laugh about the footprints on toilet seats. But I haven't seen them for years, so that's a change NEA has made!

NEA has created a culture of safety, and that's really important. We went through this with the whole SARS incident in 2003, and NEA helped create the awareness for clean restrooms and personal hygiene, like not sneezing on other people.

Here's an example of how the culture has changed. When I first came to Singapore, I remember talking to people from one of these steakhouses. They were famous for their all-you-can-eat salad bar. They said, "Look, it's not working in Singapore because no one wants to eat at the salad bar." You didn't eat raw vegetables in Asia 15, 20 years ago, because you didn't know if it was safe to do so.

People can say having these requirements increase the cost, but they also lower the cost because there is more trust across the system. I don't have to go out and seek suppliers and make sure independently that they are certified to be safe. Having this system means that when you bring somebody on board the food industry, you know they have been through a course on food hygiene. It sets a culture where if somebody gets off a boat and tries to sell you a bunch of oysters, without any licences or certifications, you say, 'I don't know where these came from!'



WE HAVE GROWN TO LOVE CLEANLINESS

Veronica Choo runs the popular Hwa Yuen Porridge at the upgraded Tiong Bahru Food Market. She took over the stall from her parents-in-law, who started out as street hawkers peddling congee along Kim Pong Road many years ago, before they were resettled here. She tells us about food hygiene in the past and today.



What would you say is the biggest difference between the old Tiong Bahru Market and the upgraded version?

Before upgrading, people who came here to eat had to put up with rain and shine. Now, they don't have to worry about rain anymore! It is also much cleaner, better now. Customers can tell at one glance if our stalls are clean. Especially tourists or locals who are here for the first time, they would tell us, it is 'very clean!' And they don't have to worry that if they take hawker food, they may end up with food poisoning. Even though some customers say they miss the old hawker centre and the old atmosphere, in truth, they won't accept it if the surroundings are not clean. It's just talk. In the past, before the upgrading, they would say, 'Why is it that the hawker centre hasn't been upgraded yet, why is it still so old?'

As someone who is a veteran hawker, do you feel that food hygiene practices have changed over the years?

In the past, our expectations of hygiene were not so high! But as we learn more about hygiene and health, our expectations have grown and now if we see poor handling of food, we won't dare to eat. We also train our stall helpers to be hygienic and send them for courses. In the past, it was so hard to make them understand the need for hygiene! We would tell them, 'Don't use your bare hands!' And they would still do it once our backs were turned. But nowadays they won't – they will use gloves or tongs. So practices have improved.

The stall holders in the past were of the older generation, so they didn't have the opportunity to be educated. Nowadays, the stall helpers and holders are more educated, so their mindsets have changed, because there is more knowledge and awareness. Our lifestyle has changed, we have grown to love cleanliness and hygiene more.

Has the smoking ban affected your business in any way?

Now, after the smoking ban, things are much better. The centre used to have so much smoke, and as stall holders, we took in so much second-hand smoke! It's a good move, because those who are sensitive to smoke can now eat in peace at the hawker centre, whereas in the past, they would have to avoid the smokers.

Do you feel that NEA's grading and inspection of food stalls is important?

Whether it is a good grading or not, at the end of the day, what really matters is that we practise cleanliness. Today, most customers expect stalls to have at least 'B' or 'A'. In the past, few stall holders would think this way, they would say, 'It doesn't matter, we grew up on unhygienic food anyway!' But now, nobody talks like that.

I feel quite relaxed when the NEA officers come by to check. Because I know all our things are in order. Such inspection has a role – it serves to remind us that we need to be clean and hygienic at all times. It's a good thing, otherwise, some stall holders may just do a superficial job. In the past, some stall helpers even smoked and handled ingredients at the same time!

Controlling Outbreaks

The duty to protect Singapore from outbreaks of vector and environment-related diseases is shouldered by NEA. The agency sees to the first line of defence for Singapore, through its comprehensive programmes of preventive surveillance and control, inter-agency collaboration, public education, community engagement and scientific research.

Vector: a living organism which carries a pathogen that is capable of causing an infection in humans.

Pathogen: a microbe or micro-organism which can cause an infection to an animal or plant.

Cockroaches, rodents, flies and mosquitoes – these are some of the common vectors found in Singapore today. Of these vectors, the mosquito stands out as the most dreaded. Unlike other vectors, the female mosquito actively seeks human prey. Through its bite, it transmits viruses or parasites it has picked up from an infected person to the victim. The mosquito is responsible for spreading pathogens causing diseases such as malaria, dengue fever, chikungunya and Japanese encephalitis. Today, the *Aedes* mosquito is Singapore's biggest concern as it spreads dengue, an endemic disease in tropical and sub-tropical areas. Effective vaccines or anti-viral drugs for the disease have yet to be developed, and those who contract dengue may face death.

NEA is Singapore's first line of defence against vectors and the pathogens they carry. The island's status as a travel hub makes it vulnerable to imported pathogens from other regions while its high population density means that multiple persons can be infected within a short time. Through its comprehensive programmes of preventive surveillance and control, inter-agency collaboration, public education, community engagement and scientific research, NEA strives to prevent outbreaks of vector and environment-related diseases in Singapore.

COMBATING DENGUE

Look around, are you breeding danger? It's your life, it's your fight. Change water on alternate days. What if unwanted guests move into your house while you are away? Stop breeding danger. These constant warnings on the radio, television, papers and posters are part of NEA's strategies to battle dengue.

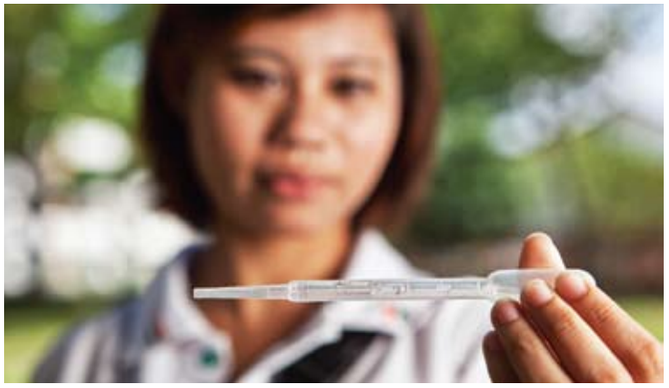
Fighting dengue is a formidable task. The *Aedes* mosquito, which has a lifespan of about two weeks, can breed in a spot of water as small as a twenty-cent coin. The female mosquito can lay about 300 eggs in its lifetime. These can hatch into larvae in less than a day, and grow into mosquitoes in a week. When a mosquito bites a person bearing the dengue virus, it becomes infective in about seven days. In short, they breed fast, infect fast.

To fight them, NEA employs two primary operational methods. One, look for signs of breeding and make the environment inhospitable. Two, break transmission of the disease as soon as dengue cases are detected.

Search and destroy. This is the mission of NEA officers. They scour public areas and residences to hunt for breeding grounds, remove stagnant water and destroy the sites. Property owners found to have mosquitoes breeding on their premises may be penalised for creating public health risks. In an average year, NEA personnel carry out at least one million inspections and ground surveys. When warmer months come around from April to September, dengue cases often spike as *Aedes* mosquitoes breed rapidly. Hence, NEA strikes in advance, launching operations aptly termed "Intensive



The Mosquito Larval Inspection Pole allows NEA officers to check roof gutters for mosquito breeding without having to resort to using ladders.



Source Reduction Exercises" in public and residential areas. These operations seek and annihilate not just breeding sites, but also potential habitats.

The war against dengue is a national issue. NEA leads the government's task force comprising some 25 agencies and institutions in anti-dengue efforts. It also coordinates and works with groups including town councils, construction sites and schools to eradicate mosquito breeding on their premises, and help them develop stringent regimes to prevent breeding. Such collaboration leads to visible results. For example, in 1999, about 30% of construction sites were found to be harbouring mosquitoes. By the mid-2000s, the figure had dropped to about 6%.

But even with all the resources pumped into this fight against the *Aedes* mosquito, it will always be an uphill battle as long as individuals do not exercise vigilance at their end. Research statistics show that mosquito breeding takes place mostly within homes. The public thus holds the all-important key to the success of these efforts. On NEA's part, it does its best to encourage every person to play his or her role. Mass media is exploited to spread anti-dengue messages via every communication channel possible. Programmes are developed to target specific groups of people, such as outbound travellers, domestic helpers and homemakers. Grassroots organisations, community volunteers, government agencies and educational institutions are also involved in dengue prevention efforts on the ground. In 2011, NEA launched a new engagement plan. This plan incorporates more in-depth education and outreach efforts with routine inspection visits by environmental health officers. During these visits, which take place before the warmer months, no enforcement action will be taken against any mosquito breeding detected. Instead, the focus is on public education.

TACKLING VERMIN AND NUISANCES

Besides mosquitoes, NEA also tackles vermin and nuisances in Singapore. It conducts regular checks for breeding and harbourage sites of cockroaches and rodents and educates stakeholders such as town councils, landlords and food operators on good housekeeping practices, refuse management and pest control. It enforces against those who repeatedly exhibit poor housekeeping. NEA works with pest control companies as well to reduce the number of rodent burrows, exterminate vermin and keep the crow population low.

MANNING THE GATES

Singapore is one of the busiest ports in the world. In 2011, more than 120,000 shipping vessels arrived at our ports while 46.5 million passengers passed through Changi Airport. Such high traffic also means that the risk of infectious diseases from other regions entering Singapore is higher. NEA holds the fort against potential outbreaks through its port health duties.

At Changi Airport, NEA officers, together with Immigration & Checkpoints Authority personnel, screen passenger arrivals. They watch out in particular for those who may be infected with yellow fever. Yellow fever is caused by a virus spread by mosquitoes and may lead to internal bleeding and death. To prevent the virus from gaining a foothold in Singapore, passengers from countries where the disease is endemic must carry valid health certificates. Those who are not certified will be examined medically and issued with a health alert card.

At the sea port, NEA officers keep a look out to prevent plague from entering Singapore. Plague is a deadly and fast-spreading disease caused by bacteria borne and spread by rats. In this case, NEA inspects vessels for signs of rat infestation. Ships which have death or a sick person on board, or which are coming in from plague-infected countries are quarantined before they are cleared to enter the port.

FROM HOUSE TO HOUSE



It was just a small note of appreciation, but it made Shaik Amina Bee's day. A family had written in to NEA to commend her for her professionalism and helpfulness. That would not have been so surprising if not for the fact that Shaik Amina had actually issued a summons to this family for breeding mosquitoes on their premises.

Shaik Amina Bee has been working as an Environmental Health Officer since 2005. She is one of the many NEA officers who go on foot, house to house, to check for mosquito breeding as part of the agency's war against dengue outbreaks. "I entered the flat and saw four huge porcelain vases on the family's altar," Shaik Amina recounts the incident. "So I informed the owners that I needed to check those vases for mosquito larvae. I helped them carry the vases and pour the water out and found larvae in one of the vases." What impressed the residents despite having been penalised was Shaik Amina's helpfulness. Not only did she help clean the vases and water spillage, she also shared information and tips on preventing mosquito breeding.

To Shaik Amina, this is all in a day's work. From 7.30 am to 5 pm every day, Shaik Amina visits residences in the blocks assigned to her. When she enters a flat, she checks water receptacles, dish trays, overturned pails, flower pots – anything that may be a breeding spot. If the resident is not at home, she has to return on another day. During dengue cluster outbreaks, gaining access to every unit is all the more important. Shaik Amina, like other environmental health officers, has to work nights to break the source of transmission. When she finds mosquito larvae on the premises, she will inform the resident, point out the larvae and collect them in front of the resident. With her educational kit in hand, Shaik Amina also educates the resident on how to prevent mosquito breeding. The larvae are then sent back to NEA for analysis.

Running into disgruntled or even angry residents is a common occurrence in Shaik Amina's work. She shares, "Sometimes, when I find breeding in the homes, the residents get angry, and they will ask, 'How come there are mosquitoes? I change the water regularly!' I hear all kinds of accusations. Some bang their doors, or complain that I'm rude. Some even snatch the dish tray and throw the water away before I can collect the larvae!"

Such unpleasant encounters only strengthen Shaik Amina's resolve to do her job well. She puts it this way, "I am trying to do my job and prevent dengue outbreaks, but some residents only think about fines. So I will hear them out, and try to explain to them the importance. Sometimes I joke and banter with them, and they will become calmer after a while. Some would say, 'Give chance *la*, officer.' And I have to tell them it's not my decision to make." She learns from these encounters too. Now, when she finds larvae, she takes a picture of them in front of the residents, so as to deter residents from tampering with the evidence.

What really fuels her is her personal conviction in the work she does. Recalls Shaik Amina, "Last time, there were mosquitoes in my flat, but I didn't even know they were mosquito larvae. I thought they were worms. So after joining NEA, I realised, this means a lot of other residents are also unaware. When I point out larvae to the residents, sometimes they will make sounds of disgust, and say 'Aren't those worms?' 'No', I say, 'Those are mosquito larvae and they are very dangerous. You can die from it.' See, a lot of people do not know what larvae are, or what dengue is. So my job is important, to educate them, and explain to them. And this is my job and I represent NEA, so I must do it well."

"When I point larvae out to the residents, sometimes they will make sounds of disgust, and say 'Aren't those worms?' 'No', I say, 'Those are mosquito larvae and they are very dangerous. You can die from it.' See, a lot of people do not know what larvae are, or what dengue is. So my job is important, to educate them, and explain to them. And this is my job and I represent NEA, so I must do it well."

– Shaik Amina Bee, Environmental Health Officer

CRUNCHING DATA TO DETECT CLUSTERS

You may have noticed this in your last encounter with an NEA environmental health officer – that he or she is tapping away at a palm device.

This small device, a Personal Digital Assistant (PDA), has greatly increased productivity for NEA's ground officers. No longer do they need to shuttle to and fro the regional offices as much as in the past. Feedback or complaints sent to NEA's headquarters are transmitted directly to ground staff, who can then respond straightaway. Enforcement can be carried out on the spot as well. Ground officers can also retrieve relevant data on the cases they are working on while out in the field, such as the location of mosquito breeding sites. But the real beauty of this gadget lies in the fact that the data keyed in by these officers is directly transmitted to a powerful system at the agency's headquarters, and used for in-depth analysis.

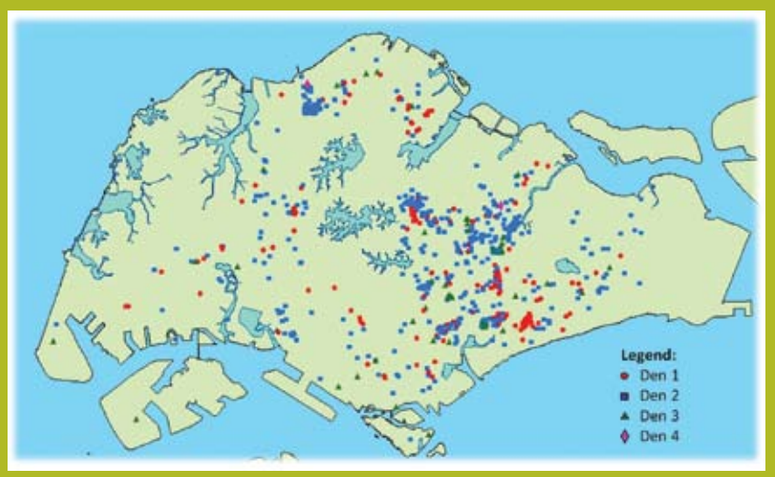
Every day, large amounts of data are created as NEA officers go on routine surveillance – the latest mosquito breeding site found, cases of non-compliance at hawker stalls etc. This data is updated real-time into a system which uses Geographic Information System (GIS) technology and internet mapping to plot the information. Relevant data from other agencies, such as SLA and MOH, is also imported. Analysis is carried out and ground operations are then planned and strategised accordingly.

Since the mid-2000s, this system has played a critical role in the fight against dengue outbreaks in Singapore. When data on breeding sites is keyed into the system, these locations show up on a virtual map of Singapore. Reported cases of dengue from MOH are plotted as well. When two confirmed dengue cases with onset dates within 14 days of each other occur in a radius of 150 metres, the system will flag it out as a dengue cluster. Other parameters are also used to pinpoint areas where there is localised transmission of dengue. Analysts at NEA can also see how dengue cases are spreading over time.

The data is updated in real-time. This allows officers at both the headquarters and regional offices to be in tune with what is happening on the ground. Using the data, they can plan operations and deploy resources more effectively. For example, the mapping system also provides information such as the locations of nearby construction sites and vacant plots of land, housing type and whether there are vulnerable groups such as nurseries or homes for the elderly in the vicinity. With the information, officers can identify which areas may be potential breeding sites and which sites should be inspected first. If the housing in the area is predominantly landed, ground staff can also prepare the necessary equipment to check roof gutters, a common breeding spot in landed property.

On top of this, NEA also takes pre-emptive measures once potential clusters are identified. Clinics located near these clusters are highlighted in the system, and NEA will inform them to watch out for dengue cases and notify MOH immediately when there are suspected cases. Early detection can help NEA to zoom in on problematic areas on the ground more quickly. When the cluster of transmission is broken at an early stage, the outbreak can then be controlled more successfully.





THE SITUATION ROOM

The Situation Room is where NEA's strategians convene, get the lowdown on the ground situation, plan and coordinate NEA's operations to tackle environmental and public health incidents such as dengue outbreaks.

The forerunner of this idea was a dedicated room set up in 1994 in the Vector Control and Research Department under ENV. The room was installed with the Intergraph CAD/CAM system to track the migration of the *Aedes* mosquitoes. GIS mapping technology was later introduced in 1997 to enhance the tracking system.

In 2003, shortly after NEA's formation, the entire GIS system was moved to a room situated on the 21st floor, next to the officer of the Director-General of Public Health. The room was named the "Situation Room". Not long after that, all five regional offices of NEA were connected to the main Situation Room at the headquarters and each regional office had a similar Situation Room. The ground data enabled the regional offices to monitor closely the state of the environment within their respective area of operation.

Over the years, the information captured in the Situation Room has played no small role in NEA's swift responses to incidents such as dengue outbreaks. As environmental issues become increasingly complex, such a function is all the more pivotal. To enhance NEA's situational awareness further, plans are now in place to construct a more powerful and technologically advanced command centre in the near future.

"NEA was formed in 2002, during a period that saw new and emerging diseases, some of them impacting Singapore. In 1998, there was an outbreak of Nipah Virus which took the lives of a number of our abattoir workers. Then, in 2000, Singapore had its first major outbreak of Hand, Foot and Mouth Disease, a normally mild childhood disease, where we saw the Enterovirus 71 turned aggressive. A year after NEA was formed, we had SARS. One of the key concerns then was to ensure and maintain a high standard of hygiene and sanitation to prevent outbreaks of infectious diseases that could endanger lives and create an adverse impact on our economy.

I read an article about how wild elephants were tracked in Africa using GIS. That was sometime in 1994 and GIS was something new then. So, I thought we could use the same idea to track the movement of the Aedes mosquito. We started using the GIS on a modest scale, using the Intergraph CAD/CAM system and upgraded it into a Global Positioning System later. I named the room where the information was centralised and displayed 'Situation Room', after a similar room in the White House in the United States. Each morning, I would go to the Situation Room to feel the pulse of what was happening on the ground. We would also hold meetings in the Situation Room, especially when we saw a rising trend in dengue or some other communicable disease, and those at the meeting could visualise on the screen what the ground situation was at that time."

— Daniel Wang, who was appointed Commissioner of Public Health in 1979 and went on to become the Director-General of Public Health in NEA (2002 - 2003)



SWIMMING POOLS AND COOLING TOWERS

Public swimming pools and buildings with cooling towers or central air-conditioning are monitored by NEA to prevent outbreaks of environment-related diseases. In the case of buildings with cooling towers, NEA requires owners to carry out regular maintenance. This is to prevent the growth of *Legionella*, a bacterium which can cause fatal respiratory infection. Public swimming pools are monitored as they are used by many people, which means that water-borne diseases may spread easily. Every month, pool operators must send samples of pool water for testing by accredited laboratories and submit the results to NEA. The agency also carries out random sampling of the water quality in public pools and inspections are carried out annually.

Scientists For Public Health

"We have been able to keep the overall incidence of vector-borne diseases low, comparable to that seen in developed countries. Take malaria for example, we knew that Anopheles *mosquitoes would normally breed in areas with stagnant brackish water. So we built anti-malaria drains in these areas over the years. This strategy has been a success ... Of course, we are fully aware that this does not mean that Singapore will not have any more cases. Located in a tropical region, malaria will be a perennial challenge. However, the public can take comfort that should a case occur in Singapore again, we have in place a system to detect the source and to swiftly stop the spread of the disease ... Dengue fever is another major challenge. Although the mosquito population has been kept low since the 1970s, dengue remains a concern in Singapore and globally ... So far, we have been able to control dengue in Singapore quite effectively. Even though we are located in a region that is highly endemic, we still manage to keep the incidence to the lowest in the tropic ... mosquitoes are living things. Like all living things, they are able to adapt and overcome our control measures over a period of time. So what we know about their behaviour 10 years ago may not be valid today. Likewise, what we know about them today may not be so 10 years down the road. It is an on-going challenge to keep up with their adaptability. This is why ENV* (present-day MEWR) *is stepping up our investment in applied research, so as to tackle these challenges posed by viruses and their vectors in our effort to stay on top of the changing situation. Today, I am pleased to officiate at the launch of the Environmental Health Institute, EHI in short."*

— Lim Swee Say, Minister for the Environment (2002 - 2004), at the launch of the Environment Health Institute, 25 April 2002



Over at the laboratories in the Environmental Health Institute (EHI), researchers race against time in an invisible war against micro-organisms.

Never in human history have we travelled as often, or as far, as in the 21st century. As we travel, we take along pathogens which have the potential to spread via person-to-person contact, water or vectors in whichever place we stop over.

On top of that, these pathogens evolve and adapt as they enter new territories, giving rise to new strains and reducing the effectiveness of tried and tested control measures. In the field of public health, scientists have to close ranks with ground surveillance forces to combat these pathogens.

In Singapore, scientific research on vectors and communicable diseases has been in place since colonial times, but never on the same scale as today. Recognising that the playing field has evolved significantly, ENV established EHI in 2002 to build capabilities in public health research so that Singapore is better prepared to address existing and emerging vector-borne outbreaks, particularly dengue.

Today, the institute is an integral part of NEA. Within EHI, molecular biologists, virologists, immunologists, veterinarians, epidemiologists and entomologists work together to safeguard Singapore's public and environmental health, assessing potential public health risks and applying scientific findings to formulate evidence-based measures and programmes to address these risks.



SURVEILLANCE IN THE LABORATORIES

To effectively combat potential vector-borne epidemics, we have to know how they spread and what can stop them. That requires firstly, identifying the pathogen responsible for causing the disease and the host or vector that spreads it and secondly, understanding the ecology of disease transmission.

EHI runs a laboratory virus-surveillance programme to provide early warning of outbreaks of vector-borne diseases including dengue, malaria and chikungunya. A network of some 300 general practitioners located all over Singapore supports the programme. When there is a suspected case of infection, the doctor will take a blood sample and send it for testing at EHI. If it is a case of infection, EHI proceeds to determine the genetic fingerprint of the virus.

With such data, the scientists in EHI can accomplish a number of things. Firstly, for diseases that are already endemic in Singapore, such as dengue, the data can provide early warning of outbreaks, and point to locations with higher risk. Secondly, the system enables the detection of new diseases, such as chikungunya. Thirdly, the capability enables thorough investigation of disease clusters through molecular epidemiology. For example, when malaria broke out in 2009, EHI assisted MOH in confirming locations of disease transmission. As a result, the intensity and scale of operations were more efficient and effective. To build and enable its capabilities, EHI continuously develops diagnostic capabilities through in-house research and evaluation.

Recognising that greater cross-border cooperation is needed to tackle tropical mosquito-borne diseases. EHI has established a strong network of international collaborators. There is regular exchange of information on the prevailing viruses and disease situation among researchers and authorities in different countries. These efforts contribute to greater regional preparedness for outbreaks and provide Singapore with an important line of defence against vector-borne diseases.



THE DENGUE VIRUS

The dengue virus has four serotypes, all four of which are circulating in Singapore. Serotypes are variants of the virus classified by the antigens found on the surface of the cell. An antigen is a substance that causes the human body to produce antibodies against it. Certain serotypes may be more virulent or deadly than others. Infection with one serotype confers immunity to the same type but not to the others.

Usually, there is one predominant serotype which is circulating. Over the years, EHI has established that a change in the predominant serotype is associated with the occurrence of outbreaks. Thus, monitoring when a switch takes place provides valuable early warning of impending outbreaks, allowing prompt vector control intervention.

EHI has also found that the dengue virus is genetically diverse, and there is a constant introduction of new strains. What this can mean is that the pressure exerted by our intensive vector control actually selects the 'fitter' viruses that can be more persistent and have a higher potential to cause outbreaks. As with serotypes, monitoring the genetic signature of viruses can also provide an additional early warning for imminent outbreaks.



BUILDING DOSSIERS ON VECTORS

Besides pinning down the pathogens causing outbreaks, EHI also builds up information on the vectors which spread them. Knowing the vectors inside out – their distribution, diversity, behaviour, life cycle and breeding habits – can help NEA decide on the best methods to eliminate specific vectors and assess potential risks of emerging or new vector-borne diseases. For instance, the breeding grounds of the *Anopheles* mosquito, carrier of malarial parasites, are usually found in parks, woods and forests instead of residences. *Aedes aegypti* which spreads dengue is more commonly found in urbanised areas. Hence, an operation targeting malaria will include rural parts of Singapore in its search and kill mission, while an anti-dengue operation will take place mainly in residential and built-up areas.

EHI's continual research has shed much light on the understanding of vectors. For example, it has developed molecular identification tools to characterise the genetic diversity of *Aedes aegypti* population in Singapore. Using them, scientists discovered that the local population of *Aedes aegypti* is genetically diverse. More research can then be conducted on the growth and distribution of the mosquitoes and how competent they are as vectors.

Another example is the institute's research on the flight range of dengue vectors – *Aedes aegypti* and *Aedes albopictus*. Prior to this research, it was commonly held among scientists that *Aedes aegypti* seldom flies more than 50 m in its lifetime. A field study by EHI debunked that belief, after it demonstrated that both *Aedes aegypti* and *Aedes albopictus* can easily and quickly disperse within a radius of 320 m to search for breeding spots. EHI also showed that mosquitoes can effectively disperse in disused high-rise residential buildings. This study led to a revolution in the understanding of Aedes mosquitoes' flight ability and won a Royal Entomological Society Award from the United Kingdom for the year 2004 - 2005. Subsequent anti-dengue operations were revised to cover a wider area.

FINDING APPLICATIONS

On top of its role in supporting ground surveillance and vector control programmes, EHI also applies its research to find practical solutions for public health issues.

Take insecticides, for example. EHI checks the susceptibility of mosquitoes to various types of insecticides so that it can identify if any of the mosquitoes has developed resistance, and which insecticides work best against them. With this understanding, NEA can ensure that insecticides used in mosquito control remain effective.

The institute has also developed a number of interesting applications. A cost-effective protocol for early diagnosis of dengue and serotyping using polymerase chain reaction (PCR) technology, for example, was developed by EHI. This protocol is now used in several laboratories in local hospitals to shorten the time taken to diagnose dengue. It also facilitates more real-time monitoring of serotypes.

Another important innovation of EHI is a non-invasive method to diagnose dengue. Traditionally, patients need to undergo a blood test to confirm an infection. Now, the institute has developed a method which can detect the infection with the saliva of the patient. The test, which is currently undergoing field trials in a few countries, is much easier to perform, more cost-effective and can be used in laboratories without the need for sophisticated equipment.

Mosquito identification tools are also developed by EHI's scientists for education, reference and training purposes. These tools include pictorial booklets and charts featuring common mosquito species and online interactive keys.

NOT JUST VECTORS

Besides vectors, the institute also researches environmental health concerns and food hygiene, such as the mould problem in indoor environments, risk of carbon monoxide accumulation in restaurants with individual cooking stoves and *Legionella* bacterial growth in spas, water storage tanks and mist fans. EHI is also developing the capability to assess the microbiological quality of local retail food, identify the risks associated in the preparation and handling of food and devise ways to mitigate these risks. All these help to ensure that our food will always be safe for consumption.





CURBING CHIKUNGUNYA

The story of how Singapore curbed the spread of chikungunya in 2008 is one which shows how inter-agency vector control operations and laboratory surveillance came together to overcome the outbreak.

Chikungunya is a debilitating viral disease transmitted by mosquitoes. As no vaccine is currently available to prevent infection, the control of vectors is critical in the fight against the disease.

Even before chikungunya reached our shores, EHI had been collaborating with relevant institutions in Sri Lanka on chikungunya research. This offered EHI first-hand experience in handling and analysing the virus, as well as a good understanding of the epidemiology of the disease.

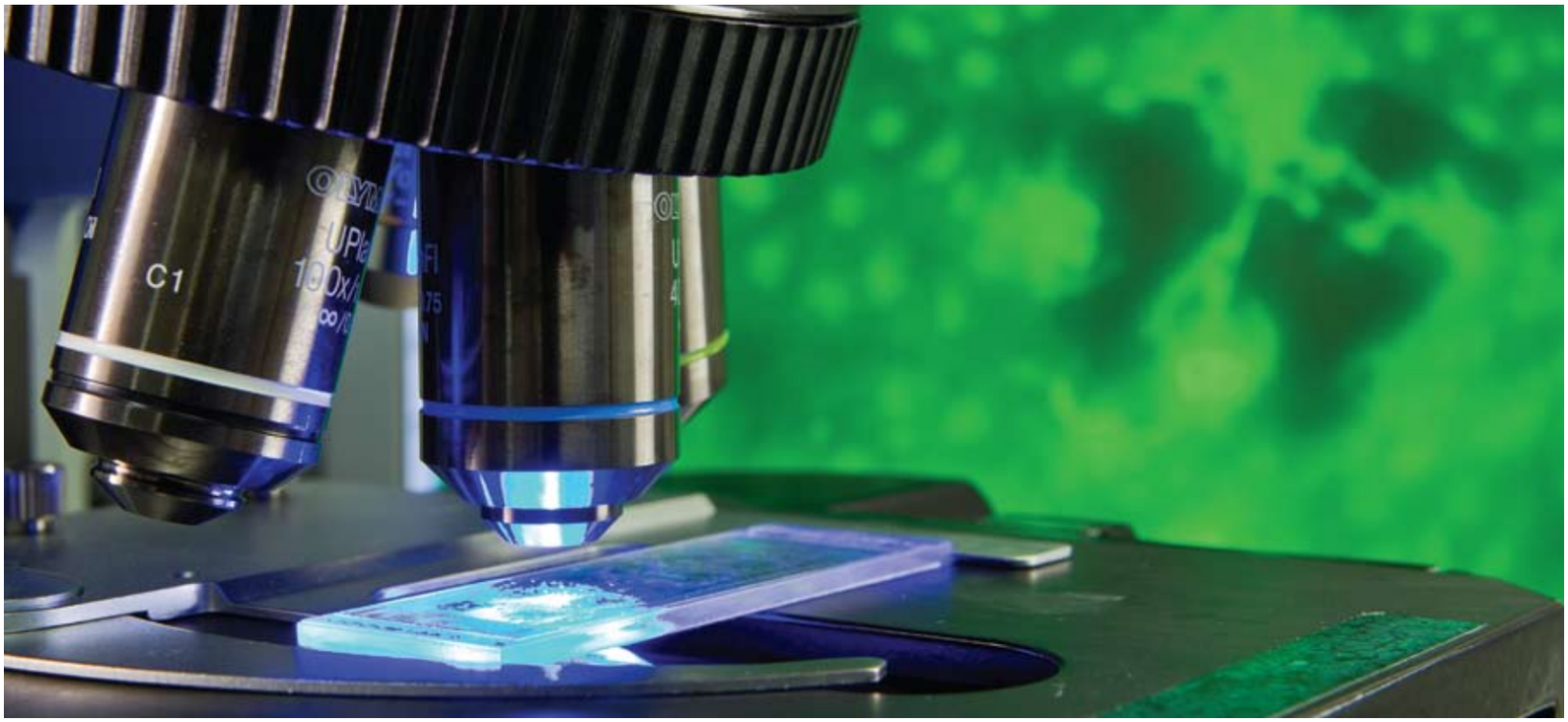
Through EHI's surveillance network, doctors in participating clinics had been alerted on the possibility of chikungunya hitting Singapore. When the disease broke out in January 2008, it was detected swiftly by EHI. Active case detection was initiated by MOH to uncover secondary cases among residents and workers in Little India and Serangoon, the vicinity of the index case. On its end, NEA promptly mobilised its ground forces to rein in the spread by stepping up vector control.

Over at EHI, researchers processed 400 to 500 blood samples daily, given the surge in number of suspected cases. NEA also involved other agencies such as URA, the Ministry of Manpower and Singapore Contractors Association Limited to clean up the area. Community outreach programmes in Serangoon were intensified to urge residents, shop owners and foreign workers to prevent mosquito breeding and to seek medical attention immediately if they experienced symptoms similar to chikungunya. These efforts successfully stopped the disease from spreading further and the outbreak was declared over in February 2008, with 13 reported cases.

However, in May 2008, chikungunya reared its head again and this turned out to be a substantially bigger wave of infection with cases reported in different parts of the country.

Again, the researchers at EHI worked fervently late into the night to track down the strains of viruses. What set this epidemic apart from the previous transmission was that the chikungunya virus was a variant spread by the mosquito species, *Aedes albopictus*, rather than *Aedes aegypti*. Armed with this newfound information, NEA promptly revised its ground strategy to target *Aedes albopictus*, by shifting its focus from the indoor environment, where *Aedes aegypti* thrive, to outdoor forested areas that are favoured by *Aedes albopitus*. Thermal fogging of outdoor areas was carried out and residual spraying on the external walls of buildings was conducted. Search and destroy efforts focused on eradicating outdoor breeding habitats of these mosquitoes.

These efforts finally brought about the control of the epidemic spanning close to a year. By the time the epidemic was over around mid-2009, close to a thousand cases were reported. The preparedness and response of NEA have often been upheld as a success by experts, some of whom have witnessed the disease infecting 30% to 50% of a population within six months of its onset in other places.



"As a WHO Collaborating Centre, EHI will forge closer linkages between scientific experts from different disciplines, so that we can synthesise innovative solutions to the dengue problem. EHI will also work with other international institutions to promote a regional surveillance network, to facilitate the timely exchange of surveillance information. This will improve the region's collective readiness and response to any arboviral disease outbreak in the future."

— Dr Vivian Balakrishnan, Minister for the Environment and Water Resources at the first ASEAN Dengue Day on 15 June 2011

AN INVALUABLE INSTITUTION

Since its establishment in 2002, EHI has been performing an important role in guarding Singapore's public health. But beyond that, this young institution has also made significant contributions to the world in various ways, from its research breakthroughs to the applications it has developed.

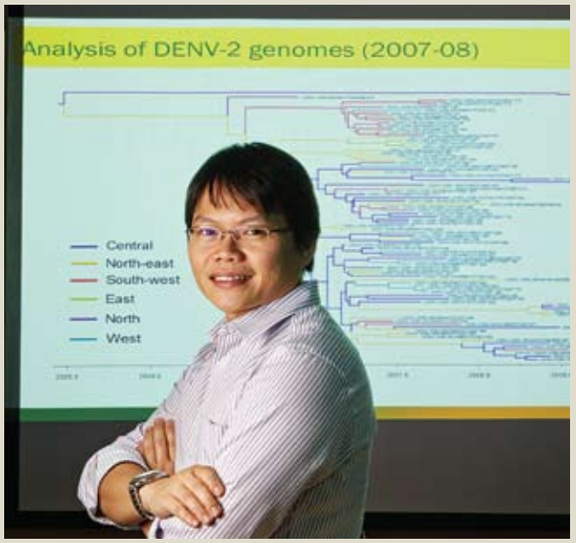
The fields of epidemiology, vector control and environmental hygiene encompass many aspects and evolve rapidly and continuously. The timely exchange of information is thus crucial to ensure that scientists worldwide can work collectively for the good of humankind.

As part of the global scientific community, EHI participates in and hosts regional and international events related to public and environmental health issues. It also partners other research institutes around the world in the quest for knowledge. For example, in 2007, EHI joined forces with the renowned Institute of Pedro Kouri from Cuba to research dengue control. Like Singapore, Cuba is afflicted with dengue outbreaks and has invested heavily in its dengue control programme. Through the collaboration, both institutions benefitted immensely from each other's expertise and experience.

EHI's research excellence is well-noted in the international community as well. Over the years, its researchers have served as advisors to WHO in meetings advocating stronger collaboration and coordination among different sectors for further global efforts against epidemics. In 2011, it was appointed by WHO to be its Collaborating Centre for the Reference and Research of Arbovirus and their Associated Vectors. As a WHO Collaborating Centre, not only can EHI strengthen its research regime, it will also play a leading role in developing a regional dengue surveillance network.

MOLECULAR MICROBIOLOGISTS AGAINST DENGUE

Dr David Lee Kim Sung, Programme Head, Research (Vector-borne Diseases) Section, is one of the molecular microbiologists behind NEA's battle against dengue.



What is it that molecular microbiologists do in EHI to combat dengue?
We sequence the virus, which helps us understand where the virus originates from and how it spreads, so that we can carry out risk assessments. The information is passed to our colleagues in vector control operations, who can then strategise how to pre-empt and manage outbreaks.

How does your work relate to dengue control in Singapore?
Through our work, we can identify which strains are more likely to cause epidemics. So when we find human cases infected by such viruses, we will intervene with our vector control to ensure they don't spread. We also find that virus strains causing outbreaks in different years and different regions can be genetically different. As the virus does not respect country boundaries, it is important for us to be aware of the situation in the region and at the same time share such information regularly with our counterparts in other countries, so that together, the region becomes stronger and more united in tackling this public health menace.

What are the challenges you face in your work?
Sometimes we have limited data available for analysis to come up with meaningful interpretations of the situation. It takes time to obtain samples from the cases, sequence and analyse them. When outbreaks occur, it is urgent to turn out the results as fast as possible, especially if they are associated with an uncommon virus strain. It's always a race against time.

What do you enjoy most about your work?
The work in EHI is very meaningful as I can contribute my knowledge and skills in science in a more practical manner, with my work having an impact on public health policies.



Setting up a mosquito trap in the course of field work.

TRACKING ZOONOSES

Grace Yap, Senior Research Officer, leads the team in EHI on the surveillance of zoonotic diseases, i.e. infectious diseases that spread from animals to humans. The team looks at diseases caused by animals that live amongst us and other zoonotic diseases that may be introduced into Singapore and have the potential to be transmitted by vectors, particularly mosquitoes. Blood samples are taken from different animals for research, including migratory birds. These samples are screened for different pathogens. One of the viruses the team watches out for is the West Nile virus, which is carried by birds and mosquitoes, both of which are found in Singapore.



PORING OVER EVERY PART

Chatting with Lam-Phua Sai Gek about mosquitoes is like stepping into a different world. Most people see nothing but an annoying insect. Not Lam. "This," she points to a photo in a booklet, "is what we call a Gigantic Mosquito. It is very big and its body is very shiny. Its larvae can feed on other mosquito larvae. And this small one here," she points again, "emits a kind of luminescence. It is so beautiful. When I look at the mosquito under the microscope, they are all different. They have different colours, different features, so you can feel their beauty. Not all mosquitoes bite people and not all are vectors."

Lam, who has been documenting mosquitoes since 1974, is a Principal Research Scientist at EHI. She is the taxonomist and chief curator of the institute's impressive collection of over 6,000 mosquitoes representing at least 90 of the 137 species known to be found in Singapore. Taxonomy encompasses the identification, nomenclature and classification of organisms. Such work is fundamental to vector control operations and research as it allows accurate and fast identification of mosquitoes.

Says Lam, "Our work is to identify the mosquitoes found in Singapore – both the adult and the larval stages. The mosquitoes are collected from different parts of Singapore, both urban and forested areas. We preserve the larva, the pupa, the exuviate (The skin shed by the adult form of the mosquito as it emerges from the pupal stage) and the adult. We catalogue all the specimens according to their species and the locations where they were found. We look at the features of the mosquitoes – the head, the thorax, every part – all using our eyes and with the help of the microscope."

This information has been put to various uses, such as developing a taxonomy key for identifying mosquitoes, training materials for ground officers and as a regional reference centre for taxonomic research. The taxonomy key, which guides users through an algorithm to determine the identity of the mosquito, is an invaluable aid in the field.

Apart from taxonomy and curation, Lam's team is also responsible for establishing and maintaining colonies of four species of mosquitoes, *Aedes aegypti* (dengue), *Aedes albopictus* (chikungunya), *Culex quinquefasciatus* (West Nile) and *Anopheles sinensis* (malaria). The mosquitoes are used for research purposes, for example, as subjects in insecticide sensitivity tests.

"We try to collect as many specimens as we can to build a database. Over the years, we have added nine new records to the documentation of mosquito species in Singapore," says Lam. "This documentation is important as it aids risk assessment. For example, the species of mosquitoes that can transmit the West Nile virus are found in Singapore too, but the disease has not occurred yet. We need to be on the alert and we have the record to show the geographic spread of these mosquitoes. So if one day, the West Nile virus shows up here, we know where to target our control."


Toxorhynchites sp.
Toxorhynchites sp. 6.5.87
Toxorhynchites sp. VCRD Lab Mar '87
Toxorhynchites sp. VCRD 2.4.98
Toxorhynchites sp.
Toxorhynchites sp. March '87
Toxorhynchites sp VCRD 2.4.98
Toxorhynchites sp VCRD 2.4.98
Toxorhynchites sp. 04 03 03
Toxorhynchites sp VCRD 2.4.98
Toxorhynchites sp. VCRD 12.8.85
Toxorhynchites sp. CP. LAB 21/3/87
Toxorhynchites sp VCRD 2.4.98

SARS: A CRISIS TO REMEMBER

"When SARS happened, everybody was living in fear. Because nobody knew what SARS was. So there were a lot of people living in fear. We mobilised all our resources. NEA was on the frontline helping to ensure public hygiene standards. The officers went out there, walked every hawker centre, every toilet, working day and night. It was non-stop operations. And the research community was trying to understand what the virus was. But they needed virus samples. At that point in time, the whole of Singapore, not a single laboratory had the facility in place to culture the virus. But our colleagues at EHI volunteered. They felt they had the process competency given their experience working on the dengue virus. So they risked their lives to handle the SARS virus, process it, prepare the samples for the research community. And they succeeded. To me, that is the display of human courage and professional commitment. Because they could have easily said, 'We too, like everybody else, were not set up to handle SARS.' But they did not. They came forward. Yes, they made a mistake post-SARs, but we should never forget the sacrifice they made during the time we were fighting SARS."

— Lim Swee Say, Minister for the Environment (2002 - 2004)

It was a crisis few would forget. In 2003, the deadly SARS (Severe Acute Respiratory Syndrome) virus broke out across the world. Singapore was not spared. Between March and May, the outbreak held sway over the island, severely impacting public health, the economy and the daily lives of residents. By 30 May 2003, when Singapore was finally declared SARS-free by WHO, a total of 238 people were infected and 33 lives taken.

NEA was barely eight months old when the crisis took place. Though it had inherited a wealth of operational systems and expertise from ENV in managing environmental health, things were still coming together on different levels. For instance, its district offices had just been restructured to form five regional offices with new operating boundaries on 1 March, the very same day SARS broke out. Ground operations and networks had yet to be fully organised, and in some areas such as public outreach and engagement, NEA was just beginning to learn the ropes.

Yet, despite these factors, NEA rose up to the call of duty. Together with the rest of the nation, the agency did the most it could to the best of its ability to see Singapore through the crisis.

TRACKING DOWN CONTACTS

When news of an atypical pneumonia case broke out, NEA was one of the first agencies to be notified due to its close working relationship with MOH on environmental health and epidemiological issues. As the infection spread, it became clear that this was no ordinary virus.

Amidst the growing urgency to consolidate and organise the clinical response to SARS, MOH also had to cope with tracking down every single person who had been in contact with SARS patients. NEA was immediately roped in to help with contact-tracing, and a number of NEA officers were assigned to coordinate this right from Environment Building.

Every day, the hospitals would gather the details of contacts to be traced and send them to MOH, who would in turn pass the list to the team of NEA officers tasked to undertake the task. This team would follow up by tracking down some 500 people per day. After the Infectious Disease Act was invoked on 24 March, NEA officers were further tasked to serve home quarantine orders, a duty that was later outsourced to CISCO. The coordination of contact-tracing was transferred to MINDEF in late April after SARS broke out in the community and the ministry's expertise was needed to track down some 2,400 people. Lam Joon Khoi, who was CEO then, remembers, "It was a national effort. NEA was supporting in some areas and taking the lead in others. We went on overdrive then, and commandeered resources from other areas, even the corporate services staff, to reinforce the environmental public health arm. It was not just part of NEA doing the fighting. Fortunately for Singapore, the contact-tracing and quarantine were working quite well. So while we were stretched, we did not reach breaking point."

ENVIRONMENTAL HEALTH – MORE CRUCIAL THAN EVER

During the crisis, environmental health and hygiene were more crucial than ever. While it was clear how deadly the virus was, information on how it spread was still incomplete. Following the first two deaths resulting from SARS on 25 and 26 March, the government took the safe approach and announced that those who died from SARS must be cremated within 24 hours, while Muslim victims had to be double-bagged in their burial.

As the government crematoria and cemetery came under NEA's management, the agency had to ensure that the deceased were handled with care to prevent any transmission. NEA was also responsible for the safe disposal of medical waste from the hospitals, including Tan Tock Seng Hospital, where the war against SARS was being waged. When an infection cluster was traced to Pasir Panjang Wholesale Market on 19 April, NEA worked swiftly with other government agencies to close down the facility. Lee Kheng Seng, who was Head of NEA's South West Regional Office (SWRO) then, recalls, "The SWRO team worked several days and nights to clean up and sanitise the Pasir Panjang Wholesale Centre. It was a multi-agency effort and the crisis brought everyone together. Following this incident, NEA also deployed officers to go round the markets and hawker centres to make sure that hawkers keep to their daily temperature self-checks as a means to prevent similar cases of infection cluster from developing."

SUPPORTING THE CLINICAL RESPONSE

At the same time clinicians were fighting against SARS in the hospitals, it was a race against time to decode the SARS virus. Yet, there were no existing



Temperature screening was a norm during the SARS outbreak in 2003.

diagnostics test, no understanding and no treatment available for the virus. MOH thus formed a consortium of research institutes and laboratories including the Defence Research Medical Institution, Tan Tock Seng Hospital, Genome Institute of Singapore and EHI. The work was distributed, with some institutions to work on diagnostics, the universities to research the pathogenicity of the virus and Genome Institute of Singapore to sequence it. EHI, which was only less than a year old, focused on cultivating the virus to facilitate the research and development.

Another important way in which EHI supported the medical community during the crisis was the processing of samples from suspected SARS cases. Former CEO, Lam Joon Khoi, highlights, "EHI was one of the few centres in Singapore then which could test if the samples were infected with SARS. They had to work round the clock, because there was a huge volume of samples. SARS symptoms were similar to flu and there were many cases of flu symptoms. The amount of effort put in by EHI staff to cope with the volume of work should not be underestimated."

SINGAPORE'S OK!

Former ENV Minister, Lim Swee Say, hit upon the idea of launching a campaign to build public confidence by raising hygiene standards in public areas. He aptly named the campaign, "Singapore's OK" (SOK). Rolled out progressively, the campaign targeted different categories of public areas – food centres, toilets, pre-schools, markets, hostels and construction sites etc. Those who volunteered to take part would be issued a decal if their premises met the criteria, so as to mark their commitment to uphold cleanliness and good hygiene practices. On 6 May, Lim Swee Say unveiled the SOK label for public toilets, the first category to be launched. The campaign received a further boost when Prime Minister Goh Chok Tong launched SOK in the community on 1 June.

Some 600 NEA officers were mobilised to visit every premises covered in the campaign, encouraging the owners to take part. The officers also worked with these premises and NGOs to promote good hygiene habits such as regular temperature checks, keeping public toilets clean and not spitting or littering. Says Dr Pang Fung Yin who was then Head of 3P Partnership in Environmental Health Department, "SARS not only resulted in the loss of lives but also affected the social and economic well-being of Singapore, and caused many people to shun public places. We were concerned with the association between disease transmission and public hygiene and responded promptly by introducing the SOK campaign. The strategy taken was one which focused on shared ownership, leveraging on the voluntary commitment and collective effort of the people, private and public sectors. It was heartening to see many grassroots organisations and owners of premises such as hawker centres, food shops, markets and pre-schools responding positively in championing the common cause of making the environment safe and healthy for all."

A SHARP LEARNING CURVE

Despite its youth, NEA provided critical support in many ways, from taking care of environmental health issues and ensuring high public hygiene standards to contributing through laboratory work and restoring public confidence.

Unfortunately, a few months after the SARS outbreak, a lapse in safety procedures at EHI led to a postgraduate research student attached at the institute to be infected by the SARS virus. Following the incident, an international expert panel appointed by ENV identified a number of shortcomings in the physical set-up and procedures at the laboratory.

The findings led EHI to suspend work that required Biosafety Level 3 facilities till its new premises at Biopolis were ready in 2006. Biosafety levels indicate the level of containment needed in the handling of biological agents, with BioSafety Level 1 as the lowest and Level 4 as the highest. In the new premises, emphasis was placed to the proper design and maintenance of the new high containment facility. EHI also set up its Biosafety management programme and appointed staff dedicated to facility management. In 2006, MOH granted approval to EHI to possess and handle biological agents in its new Biosafety Level 3 facility, which was certified to meet or even exceed MOH's prescribed standards.

Besides this sober lesson in biosafety, NEA had matured in various ways through the crisis as well. Its ground operations had been considerably strengthened and the seeds for greater community involvement in the environment had been planted. Lam Joon Khoi recaps, "The SARS crisis reinforced my view that there is a need to develop strong partnership with our key stakeholders – community leaders, NGOs, industry associations and other public agencies and such during periods where there are no crises to achieve our environmental goals. That same relationship can then be harnessed for swift and effective actions when crises do emerge. Coming out of the SARS crisis, there was greater public awareness on the importance of good environmental health, increased appreciation of NEA's contributions to Singapore's environmental health by the public and a stronger sense of unity among NEA staff as they overcame the difficult situation together." These insights, in a way, prepared NEA for future crises to come.

COMING INTO ITS OWN

May 2005. Cases of dengue infection began hitting more than 100 every week. By August, the weekly figure had soared to more than 400. The rise was relentless. Hospitals and clinics were severely stressed, the public was increasingly concerned and the virus showed no signs of slowing down. By September, it had climbed to more than 700 infections a week.

In a way, this was not out of the blue. Dengue cases had been steadily climbing since 2000, hitting 9,459 cases in 2004. Research has shown that dengue transmission generally peaks every six years, and the last peak had taken place in 1998. But in 2005, the sheer scale of the infection was unexpected, with 14,006 laboratory-confirmed cases and 27 fatalities in all. It was the worst dengue outbreak in Singapore's history.

Yet it was also the event that saw NEA coming into its own in controlling outbreaks. Since the SARS crisis in 2003, NEA had been honing its operational strengths and involving other government agencies, community organisations and volunteers in its programmes. These relationships united the groups together in the fight against dengue.

DEVISING A NATIONAL STRATEGY

Things came to a head in September 2005, when the infection rate spiked. NEA had relied on its routine vector control programmes to cope with the situation. But with the outbreak reaching epidemic proportions, more had to be done.

Singapore's leadership began formulating a coordinated response on a national level. Dr Yaacob Ibrahim, then Minister for the Environment and Water Resources, combined forces with Khaw Boon Wan, then Minister for Health, to chair an inter-ministerial committee overseeing dengue control efforts. On the working level, multiple government agencies formed a task force led by NEA. The mayors of the five community districts united and established a Dengue Watch Committee to work on outreach efforts. Singapore also appointed an expert panel comprising local and international experts to advise the government on prevention and control measures. At the peak of the crisis, Prime Minister Lee Hsien Loong stepped out to front the campaign, giving it much-needed support and endorsement.

AN ALL-OUT BLITZ

It was all or nothing. The task force embarked on a nationwide blitz against mosquitoes and their breeding sites on 18 September. Mobilising more than 6,000 volunteers from government agencies and social organisations, NEA led them in carrying out "carpet combing" of public outdoor areas for six consecutive weekends. The term refers to officers going over the ground with a "fine toothed comb" to seek out every mosquito breeding site. More than 1,000 spots and 8,400 potential breeding sites were destroyed.

The war made its way into homes with the "10-Minute Mozzie Wipe-Out". This massive outreach programme encouraged the public to spend just 10 minutes a day to remove stagnant water from their residences. Some 10,000 volunteers distributed the campaign pamphlets to 880,000 homes. The message was spread to construction sites, shipyards, factories and foreign domestic helpers as well.

On its end, NEA increased its field deployment of officers from 110 to 510. The entire NEA was mobilised too and for two months, officers from other departments took part in the carpet combing exercises. More than 934,000 premises were inspected and 52,000 sites surveyed by NEA in 2005. A public hotline dedicated to dengue was set up by the agency in September, and more than 16,000 calls were received in total. Every call was investigated within 24 hours and tips on breeding sites were followed up.

TARGETING DRAINS AND ROOF GUTTERS

Through the task force, NEA mobilised government agencies and private organisations to take ownership of eradicating breeding sites on their premises. Infrastructural checks, repairs and renovation were carried out to eliminate sources of stagnant water. Cracks were sealed and land was backfilled. The frequency of checks and cleaning of public drains and roof tops was increased. For closed drains and hard-to-reach areas, misting of larvicides was carried out. Points of contact were identified in each organisation so that NEA could liaise with the right person to react swiftly in cluster outbreaks. NEA also helped the agencies evaluate their procedures to cut down mosquito breeding.

Adopting a long-term view, NEA also worked with HDB and BCA to review designs of drains and roof gutters to improve water flow. Roof gutters were banned for new developments from then on, and 10 km of public drains were repaired.

IN THE LAB

Given the speed of infection, time was of the essence both for vector control and clinical treatment. In the aftermath of the epidemic, the researchers at EHI developed a polymerase chain reaction (PCR) based diagnostic assay for dengue which cut short the time needed to diagnose the virus. This meant that doctors could detect cases more quickly and operational response on the ground could be initiated much earlier. Previously, the diagnosis could take a few weeks. The new assay gave an accurate diagnosis as early as the first day of the disease, and even determined the serotype of the virus. However, the assay was less able to detect the virus from the second week of infection. Subsequently, EHI also developed another assay which could detect infection using saliva from the patient, with a 93% sensitivity between four and eight days after the onset of fever.

BUCKING THE TREND

The intensive efforts from September onwards finally led to a slowdown in the outbreak by November 2005. As a result of the campaign, the public had become much more aware of the importance of dengue prevention. NEA had also strengthened its vector control and surveillance programmes considerably.

But it was no time to let up. Dengue was still a significant threat. NEA continued investing more resources in its programmes to control the disease. One significant transformation was the development of a system which integrates information from cases of infection, virus and vector surveillance and the ecology of mosquitoes to plan vector control operations.

In 2007, another outbreak took place. Based on the research on the serotypes responsible for the outbreak, EHI concluded that it would be massive in scale. NEA's operations immediately swung into action, and launched intensive vector control exercises. The impending outbreak was successfully curbed and dengue cases in subsequent years did not reach the same scale as in 2005. The dengue cycle had been successfully bucked.



"It was the most serious dengue outbreak the island has seen, with the highest number of dengue cases and deaths ever recorded. Fighting the outbreak required the mobilisation of the entire nation to get everyone to play his or her part to prevent mosquito breeding and break the chain of dengue infection. This was a tremendous challenge which NEA met head-on. Officers were rostered over a period of about two months including weekends to conduct "search and destroy" operations in partnership with grassroots organisations and volunteers to comb through every single public and private estate and other public areas to remove mosquito breeding grounds. No stones were left unturned, no stagnant water was left untouched."

— Lee Yuen Hee, CEO, NEA (2005 - 2009)











"They used to be called 'teh-gu, teh-gu'. They were seen to be very harsh on people, on street hawkers and so on. But if not for these environmental health officers, our environment will not be that clean. They are the ones who make sure that public health standards are kept. They are not afraid of enforcing the rules because they know they need to do so in order to safeguard the interest of the majority. Over the years, they have also changed in the way they undertake their roles on the ground; now they also educate as they enforce. Our environmental health officers should be recognised for the important role they have and will continue to play in keeping Singapore clean and safe."

– Khoo Seow Poh, Deputy Chief Executive Officer, NEA



WITH DUTY AND PRIDE

They have been called *Kerani Sampah*, *Kerani Jamban* and *Kerani Longkang**. Older generations of Singaporeans nicknamed them "*teh-gu*"**.

These unflattering terms arose during a time when tough action was needed to remedy the deplorable state of environmental health in Singapore. Environmental health officers, first under MOH, followed by ENV, were the frontline in enforcing new stringent laws. Today, these officers in NEA continue the tradition of safeguarding a clean, healthy city. With some 2,000 of them on the beat, NEA environmental health officers are almost synonymous with the agency. Many of them have been in this line for decades, some even from the days of MOH.

Their work is often onerous, unpleasant and even dangerous at times. Says Koh Char Boh, who joined this line in 1972, "I started out with collecting night soil from the *kampongs* (Malay: villages), and also worked at the Lorong Halus dumping ground. Even if we changed our clothes after work, people could still tell we came from the dump when we took the bus home."

Cracking down on illegal hawking, for instance, was fraught with risks in early days. "During my time, there were so many assault cases! Some of us were chased by hawkers with choppers, some were even slashed," recalls Mohamed Jumahin Bin Ibrahim, who started out in 1969. "There would be commotion when we arrived. The public would scold us, 'Give them chance *la*, don't disturb their rice bowl.' But we have to do our duty. When we set off for work, our families would remind us to be careful."

These same people also ensure food hygiene standards, control vector population, manage operations in the cemetery, crematoria and columbaria, and audit waste collection and cleaning services. When public calls come in requesting site visits, they are the first to respond. They know the ground intimately and often put the knowledge to good use, for example, in vector control. "We look high and low, check roof tops, drains," recounts Saliman Bin Mordi, who has been hunting mosquito breeding sites since 1968. "If we can't find the breeding spot, we look for clues. For example, if we see a bird drinking or taking a bath in some hard-to-reach area, or small plants growing on a signboard or such, there must be water nearby, and mosquitoes are bound to be there."

Their ground connections and knowledge are all the more important during crises. Ravindran Nair, who joined ENV in 1979, remembers the dengue crisis in 2005, "We had to train the volunteers from other agencies and the grassroots fast and plan the logistics. For one whole month, it was carpet combing every day."

Over time, the image of the work has evolved. Now, officers educate at the same time they enforce. Mohd Hussain Bin Yusoff joined ENV in 1975 and sees this change first-hand. He explains, "Environmental hygiene issues are very dynamic, and to keep up with them, we need to change public mindsets. We have to reach out to different groups of people including licensees and food handlers, and educate and motivate them to improve their standards."

Yet, despite the fact that their work is so essential to our lives, it is most of the time underappreciated. Unpleasant encounters with members of the public are part and parcel of life. Imran Bin Mohammed Aleh, part of the workforce since 1995, says, "Some residents scold us when we find mosquitoes breeding on their premises. But they don't really understand that when they breed mosquitoes, it will affect their family and neighbours as well."

What fuels these officers is a strong sense of duty and pride, as well as the knowledge that they are making a difference. Akbar Bessir has been in this line since 1972. He says, "Every year, we see improvement in hygiene standards. If you compare the *pasar malams* (Malay: night markets) 10 years ago and now, so much has been achieved in the operations of the stalls and observance of food hygiene. There is also less dis-amenity caused to residents, because of the efforts we put in. It's a lot of satisfaction."

Mohd Zulkefli Abdul Rahman is one of the younger members, having joined in 1995. "To be honest, our work here is very interesting and challenging. Not many people want to work in environmental health, but every time there is a public health crisis, we are the ones who enter the danger zone first, such as during the SARS and dengue outbreaks. That is why I feel proud to be in NEA."

For S Mohan, a veteran who joined in 1978, it is about camaraderie too. He says, "We take pride in our jobs, even if when the work is dirty. All of us played a small role, but we contributed to the bigger picture. This is where my big family is. My strength, my colleagues and my friends are all here."

* In Malay, these mean "Garbage Officer", "Toilet Officer", "Drain Officer".

** This Hokkien term translates to "Ox on the ground", probably to describe the chaos caused when the Special Squad arrives to bust illegal hawkers, akin to an ox running wild.



ON A DAILY BASIS

Daily-rated employees are a significant part of NEA's ground operations force. Many of them have been in this line since ENV times. They play an important role in maintaining the cleanliness of public areas in Singapore and preventing mosquito breeding on a daily basis. To help increase their productivity, enhance their personal development and improve their image, NEA took steps in 2011 to redesign their work scope, introduce new uniforms and increase the pay scale. Besides taking care of sanitation and vector control, the employees will also report on irregularities observed on the ground. This will allow NEA to respond more readily. Supervisory positions have also been added. Daily-rated employees in such positions are in charge of deploying and supervising staff and reporting on irregularities through Personal Digital Assistant devices.

Care For The Departed

NEA's work touches almost every aspect of our lives, even during the times when we lose our loved ones. This is one lesser-known aspect of NEA's work – that it oversees the day-to-day operations and maintenance of government cemeteries, crematoria and columbaria.

A NEW BURIAL SYSTEM

A key challenge faced by NEA when it comes to after-life services is the fact that Singapore is land-scarce. The practice of burials is unsustainable given our land constraints and developmental needs. In 1962, the government built the first public crematorium, located at Mount Vernon. Over time, cremation had become the preferred choice for the majority of the population.

At the same time, Singapore has to cater to those who wish to opt for land burials, such as those whose faiths require the deceased to be buried, including followers of Islam, Judaism, Zoroastrianism (Parsi), the Ahmadiyya faith and Bahá'í faith. These needs have to be met within the country's only cemetery in use, the 318-ha Choa Chu Kang Cemetery.

Choa Chu Kang Cemetery began operations in 1947. By the late 1990s, it was estimated that the cemetery could only last till 2013 based on existing land burial methods. A way to prolong the lifespan of the cemetery had to be found.

In 1998, ENV introduced a policy to limit all burials to 15 years, after which the graves will be exhumed. This policy was drafted after ENV studied similar legislation in places such as Hong Kong and Turkey. For those whose religions forbid cremation, the remains will be re-interred in smaller plots. Otherwise, the remains will be cremated and stored in a columbarium. The exhumed plots will then be used for fresh burials. This policy extended the cemetery's lifespan by about 50 years.



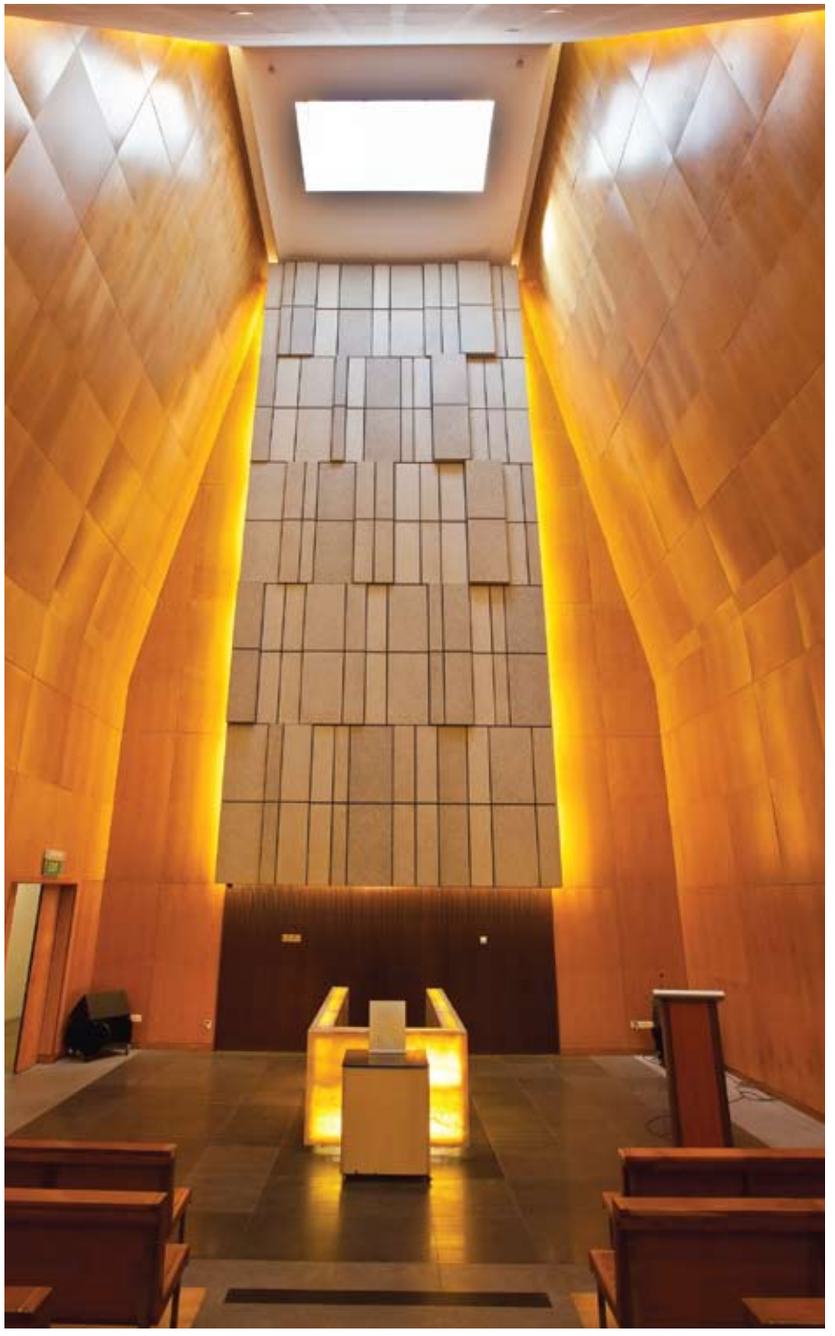


Next, the government began looking into new methods of burial. After NEA took over the management of these facilities in 2002, it continued researching how other land-scarce countries handled burials. Eventually, NEA decided to partner HDB, tapping on the board's technical expertise to develop an engineered crypt system. The system, called New Burial System, comprises prefabricated rectangular concrete boxes without bottoms, laid in a grid on the ground. Each box is fitted with a cover. When a burial takes place, the cover is lifted and the coffin or remains are lowered into the crypt, which is then backfilled with earth and covered.

There are several advantages to the new system. Manual digging of graves is no longer required. The plot density is increased significantly. In traditional land burials, graves must be spaced at least 45 cm apart for the soil to be stable. The distance is now reduced to 15 cm as the grave is separated by a thin concrete wall. Burial and exhumation processes have also become easier with the new system.

Before the system was built, consultations were held with key stakeholders including religious groups and authorities such as Majlis Ugama Islam Singapura. The system received approval and endorsement from the stakeholders and the New Burial System was implemented in 2007 in Choa Chu Kang Cemetery.

As of May 2012, a total of 23,301 crypts were constructed for burials of the recently deceased and re-interment of exhumed remains. The new burial system clinched the Institution of Engineers Singapore Prestigious Engineering Achievement Award in 2009 as it showed that engineering solutions can also be applied to traditional fields such as burials. More importantly, the system, together with the 15-year burial policy, has extended the lifespan of the cemetery to beyond 2130.



SUBTLE CONSIDERATIONS

Constructed in 2004, the planning and design of the Mandai Crematorium and Columbarium Complex actually began way back in 1994 under ENV.

Singapore being such a diverse society, the planners took care to consider specific religious and cultural needs in the design, from the use and flow of space to the type of facilities. "We consulted various religious bodies and funeral directors as we needed to understand the sequence and rituals relating to specific funeral services," explains Chow Mei Ling, who was on the team implementing the system. "For some religions such as Hinduism, believers have a custom of collecting by hand the cremated remains of their loved ones. So we built a private space for the family members to collect the ashes."

The attention to details can clearly be seen in the complex's design, from the controlled ambient temperature and staggered schedule of cremation sessions to the speed at which the coffin is conveyed to the furnace. Quiet rooms provide space for users to grieve in private. The facility's layout is intuitive with clear directional signage. Subtle features such as lighting design, landscaping and colour schemes help ease the next-of-kin's state of bereavement.

"Mandai Crematorium and Columbarium Complex was designed to cater to the needs of different religious groups, while at the same time staying non-sectarian so that it is all-encompassing. We make our considerations for religious and cultural beliefs very subtle and in the background, so that anyone who uses the facilities does not feel uncomfortable."

— Chow Mei Ling, Senior Assistant Director, NEA

Technological solutions applied at the complex are cost-effective, durable and designed to meet Singapore's long-term needs. The density of niches in the columbarium is also higher than in other columbaria to cater for future use. Says Andrew Lau, Head, Facilities, Planning and Development Section, NEA, "Even though the land use is quite intensive, you don't feel the space crunch when you are at the facility because we really took care to consider how people will feel when they visit their loved ones here. We see to it that the facility caters to the deceased as well as the living. Here, the focus is on the living's memory of their loved ones."

